

TV Advertising Effectiveness with Racial Minority Representation: Evidence from the Mortgage Market

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Abstract

This paper examines how racial minority representation in advertising influences its effectiveness. We analyze a dataset of 9.7 million mortgage refinance loans alongside data on TV advertisements for mortgage refinancing. Using the ad videos, we construct a measure of minority representation, and extract additional video and textual features using a variational autoencoder and a text embedding model. To estimate the effect of minority representation in ads on consumers' choice of lender, we apply a Double Machine Learning model, controlling for high-dimensional image and textual features as well as a rich set of fixed effects. We find that ads with higher minority representation are more effective in influencing consumer choices. Specifically, as the minority share in ads goes from 15% to 30%, the advertising elasticity increases from 0.0161 to 0.0269. This effect is more pronounced among minority borrowers and liberal-leaning borrowers. To complement our observational study, we conduct a pre-registered lab experiment where we manipulate the race of actors in a refinancing ad using generative AI technology. The experimental results are consistent with those from our observational study, providing further causal evidence. We discuss potential mechanisms underlying these effects and implications of our findings.

Keywords: minority representation; TV advertising; double machine learning; image and text embeddings; deep neural nets

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1 Introduction

In recent years, the topics of diversity, equity, and inclusion (DEI), particularly racial minority representation, have been widely discussed across various sectors of society. Efforts to promote DEI can be seen in many areas, such as inclusive hiring practices in the workplace, fostering diverse student bodies in education, and including diverse characters and narratives in media. At the same time, there are also concerns over such initiatives. Indeed, perspectives on DEI and minority representation can vary across the population.

From a marketing perspective, it is unclear whether featuring minority actors in advertising is effective. On one hand, increased minority representation may help companies better engage with their minority customer base, since consumers respond more positively to ads featuring actors from their own racial background (e.g., [Deshpandé and Stayman, 1994](#); [Aaker, Brumbaugh, and Grier, 2000](#)). On the other hand, it is unclear whether such representation may alienate White consumers. Beyond racial fit, companies may also use minority representation in ads to signal their commitment to DEI. However, the effectiveness of doing so is still an open question.

This paper investigates the impact of racial minority representation on the effectiveness of TV advertising in the empirical context of mortgage refinancing. Refinancing a mortgage is one of the most financially consequential decisions that a household can make. In 2020 alone, \$2.6 trillion worth of mortgage loans were refinanced.¹ Prior studies have shown that minority consumers are less likely to refinance their mortgages compared to White consumers with similar characteristics, forgoing substantial potential savings ([Gerardi, Lambie-Hanson, and Willen, 2021](#); [Gerardi, Willen, and Zhang, 2023](#)). The lower refinance take-up rate among minorities contributes to the well-documented racial disparities in the mortgage market (e.g., [Bartlett et al., 2022](#); [Bhutta, Hizmo, and Ringo, 2022](#)). From this perspective, understanding the impact of minority representation in ads in this market holds particular social significance.

More specifically, we aim to answer the following research questions. First, how does the effectiveness of TV advertising change with the level of minority representation? After showing the main effect that higher minority representation increases advertising effectiveness, we further investigate potential heterogeneous effects. In particular, how do the effects vary based on the borrower’s race and political ideology? Finally, taking together results from both observational and experimental studies, we discuss several potential mechanisms that are consistent with our

¹<https://www.statista.com/statistics/205946/us-refinance-mortgage-originations-since-1990/>; Accessed July 12, 2023

findings.

To answer these research questions, we obtain loan-level mortgage origination data from 2018 to 2021. This dataset includes detailed information on the borrower’s race and census tract of the property, which allows us to link the home location to census tract-level political voting data. We then merge this loan origination data with TV mortgage advertising data obtained from Vivvix (formerly Kantar Media), covering the same time period. In addition to advertising spending data, we also obtain the video files of the TV ad creatives. Using these videos, we apply computer vision techniques to determine the race of each actor and construct a measure of minority representation in advertising.

There are two main challenges in estimating the impact of minority representation on TV advertising effectiveness using observational data. First, lenders may advertise more in areas or during time periods where the expected demand is high. We address this challenge by including lender-market (Designated Market Area, or DMA) fixed effects and lender-time (year) fixed effects. Second, the level of minority representation in ads may be correlated with video features, such as visual elements or advertising messages, which can also affect the effectiveness of the ads. We address this challenge by including high-dimensional video attributes as control variables.

We incorporate both fixed effects and high-dimensional video attributes using a Double Machine Learning (Double ML) model (Chernozhukov et al., 2018). More specifically, we use a neural network with three separate branches to capture lender-market fixed effects, lender-time fixed effects, and video features. The video attributes include image embeddings, which we obtain by training a variational autoencoder (VAE), and ad transcript embeddings, which we obtain from a pre-trained embedding model from OpenAI. The Double ML estimator utilizes residual variations in both the outcome and treatment variables (ad spending and minority representation in ads) after “partialling out” the fixed effects and video features using neural networks. This approach helps isolate the key variables of interest from potential, high-dimensional confounding factors.

Our results show that increased minority representation leads to an increase in the effectiveness of ads. Specifically, we find that as the minority share in ads goes from 15% (representing the median value at the lender-DMA-year level in our data) to 30% (about a one standard deviation increase), the advertising elasticity increases from 0.0161 to 0.0269. Moreover, we observe that the impact of minority representation is more pronounced among minority borrowers than White borrowers. Across the political spectrum, we find a stronger effect among liberal-leaning consumers (at the census tract level) compared to conservative-leaning consumers. When examining interactions

between borrowers’ race and political ideology, the effect is the most pronounced for minority liberal consumers, followed by White liberal consumers and minority conservative consumers. The effect is the smallest, but still positive, for White conservative consumers.

We further complement our observational study with a pre-registered lab experiment where we directly manipulate the race of the families in advertisements using generative AI technology. The results from our experimental study align with the main effect observed in our observational study: Participants who are randomly assigned to ads featuring minority families report a higher likelihood of applying for and recommending the advertised lender compared to those randomly assigned to ads featuring White families. Furthermore, the experimental results show consistent patterns based on the participants’ self-reported race and political ideology. The impact of minority representation is stronger among minority and liberal-leaning consumers. However, compared to the observational study, we find weak evidence that minority representation might not always have a positive effect, especially among White conservative consumers.

There are several possible mechanisms that are consistent with our results. First, prior research has documented that minority consumers have a stronger preference for racial homophily than White consumers (e.g., [Deshpandé and Stayman, 1994](#); [Aaker, Brumbaugh, and Grier, 2000](#); [Mollica, Gray, and Trevino, 2003](#)), and liberal-leaning individuals have greater support for racial diversity and equity than conservatives (e.g., [Agarwal and Sen, 2022](#); [Babar, Adeli, and Burtch, 2023](#); [Aneja, Luca, and Reshef, 2025](#)). Second, through follow-up questions in the experiment, we find that ads featuring minority actors lead to more favorable brand perceptions, such as perceptions about the breadth of loan options, the fairness of the bank’s lending practices, and the firm’s inclusiveness toward individuals of all backgrounds, all of which can contribute to higher ad effectiveness. Lastly, ads featuring minority actors can be more effective simply because they are less common, making them stand out to consumers ([Pieters, Warlop, and Wedel, 2002](#); [Rosengren et al., 2020](#)). Indeed, our experimental participants rate ads featuring minority actors as being more new, fresh, and attention-grabbing than ads with only White actors.

Our findings offer valuable insights for brands when setting their advertising strategies: By featuring minority actors, brands can not only signal their commitment to DEI, but can also increase the effectiveness of their advertising efforts. Our results also have important policy implications: Given that featuring minority actors is particularly effective in reaching minority consumers, it has the potential to be a useful strategy to provide these consumers with information about refinancing opportunities and encourage them to refinance, especially in times when interest rates are low.

This, in turn, could help reduce racial disparities in the refinance take-up rate.

The two closest papers to ours that we are aware of are two concurrent working papers by [Hartmann, Netzer, and Zalta \(2023\)](#) and [Overgoor et al. \(2023\)](#). [Hartmann, Netzer, and Zalta \(2023\)](#) study online display advertising and find that online ads featuring minority actors achieve higher click-through rates than those with White actors. [Overgoor et al. \(2023\)](#) study the impact of Black actor share in TV ads on consumers’ stated purchase intentions and find that the effect depends on brands’ historical level of minority representation in ads. Our study differs from these sets of papers in a few important ways. First, we study advertising effectiveness with actual consumer purchasing data rather than relying solely on clicking behaviors or self-reported purchase intentions. Second, we account for the potential correlation between the race of the actors and visual and text features in ads in order to minimize omitted variable biases. Thus, we have a much richer set of controls in our study. Third, we leverage information on consumers’ race and political leaning to estimate heterogeneous effects along these dimensions. These differences, as well as our experimental results, also allow us to discuss potential mechanisms behind our findings.

Our research also contributes to several broader streams of literature. First, our paper is closely related to the growing literature on the impact of DEI initiatives and social equity movements on both society and business. In terms of societal impact, [Agarwal and Sen \(2022\)](#) find a significant increase in demand for anti-racist books requested by public school teachers following the killing of George Floyd. From a managerial perspective, [Balakrishnan, Nam, and Buell \(2023\)](#) and [Khan and Kalra \(2022\)](#) demonstrate that signals of diversity at the corporate level have a positive impact on consumer attitudes. Furthermore, signaling racial identity can increase demand for minority-owned businesses on platforms like Yelp and improve the success rate of requests for help ([Babar, Adeli, and Burtch, 2023](#); [Kirgios et al., 2022](#); [Aneja, Luca, and Reshef, 2025](#)). Beyond racial diversity, [Goli and Mummalaneni \(2023\)](#) find that increasing the screen time of women positively impacts the viewership of cable news shows, while [Khan et al. \(2024\)](#) show that featuring diverse models in ads (across race, gender, and age) increases brand outcomes in lab studies. However, it is important to note that DEI initiatives may not always receive favorable responses, underscoring the need for careful assessment of the potential benefits and risks associated with such initiatives. For example, [Wang et al. \(2022\)](#) find that firms’ social media posts related to the Black Lives Matter movement reduce consumer engagement on social media platforms. Additionally, recent political pressure and public debate on this topic, not covered in the time period of our paper, also suggest that views towards DEI may be malleable and may shift over time. Our paper contributes to this literature

by studying how minority representation in TV ads impacts advertising effectiveness.

Our paper also adds to the literature on the impact of advertising content. Beyond lab experiments, ad content has received less attention than the examination of ad quantity in economics and marketing. [Bertrand et al. \(2010\)](#) measure the effect of informational content (e.g., interest rates) and non-informational content (e.g., a photo featuring an attractive woman) in direct mail ads for consumer loans through a large-scale field experiment, and find that ad content has a strong role in the effectiveness of an ad. Since then, marketers have utilized either field experiments (e.g., [Sudhir, Roy, and Cherian, 2016](#); [Sahni, Wheeler, and Chintagunta, 2018](#); [Morozov and Tuchman, 2024](#)) or observational data (e.g., [Liaukonyte, Teixeira, and Wilbur 2015](#); [Lee, Hosanagar, and Nair 2018](#); [Tsai and Honka 2021](#); [Fossen et al. 2022](#)) to gain deeper insights into the impacts of both informational and non-informational ad content. Building on this literature, our study explores how racial representation in ads, as a form of non-informational content, can influence consumer demand for the advertised lender.

The rest of the paper is organized as follows. [Section 2](#) describes our main data and provides descriptive statistics. [Section 3](#) describes the features we extract from the video data. [Section 4](#) presents our empirical strategy. [Section 5](#) documents the empirical results. [Section 6](#) describes the online experiment and documents the experimental results. [Section 7](#) discusses potential mechanisms. Finally, [Section 8](#) concludes.

2 Data Source and Descriptive Statistics

In this section, we describe the two main datasets used in our study: the mortgage origination data ([Section 2.1](#)) and the TV advertising data ([Section 2.2](#)). We discuss how we extract race information from the advertising video files and present descriptive statistics in [Section 2.3](#). Finally, we present the descriptive statistics for the estimation sample in [2.4](#).

2.1 Mortgage Refinance and HMDA Data

Our analysis focuses on conforming mortgage loans. In the United States, the vast majority of mortgages are conventional loans that are not directly backed by the government, unlike FHA and VA loans, which are originated for borrowers with lower credit scores and for active-duty service members and veterans, respectively. Within conventional loans, we focus on conforming loans,

which are originated by private lenders, such as banks or mortgage companies, but follow the guidelines set by government-sponsored entities (GSEs), Fannie Mae and Freddie Mac. These guidelines include credit score thresholds, loan size limits, and debt-to-income ratios. Most conventional loans are conforming loans, as lenders typically sell conforming loans to GSEs within a few months of origination to free up capital for new lending and transfer default risk to GSEs.² Due to these institutional details, lenders tend to use similar underwriting criteria for conforming loans.

Since the vast majority of mortgage loans in the U.S. market have a fixed rate, refinancing a mortgage when market interest rates are low is the primary way for borrowers to take advantage of the lower rates. While borrowers with adjustable-rate mortgages will see their interest rates drop when the market rates decline, many of these borrowers may also want to refinance and switch to a fixed-rate mortgage that locks in the low rate. Either way, when a borrower refinances, they pay off the existing mortgage (i.e., prepayment) and take out a new loan at the new rate, all in one action. Some borrowers may also choose a cash-out refinance, which allows them to get a loan for a higher amount than the existing mortgage based on their increased home equity, generally for a slightly higher interest rate than a non-cash-out refinance.³

Refinancing is a common action and accounts for a large share of the total mortgage market. According to data from the Mortgage Bankers Association, refinancing made up 53.67% of total loan origination volume during our sample period, although this varied significantly across years depending on the interest rates. For our years, the range was from 28% in 2018 to 64% in 2021.⁴

Borrowers may refinance with either their current lender or a new lender. While we do not observe consumers' previous lenders in our data, an industry report finds a retention rate of about 20% for lenders in this market,⁵ suggesting that switching lenders is fairly common. We also observe that borrowers tend not to shop around extensively to find the best rate.⁶ Using publicly available National Survey of Mortgage Origination data, which is a quarterly mail survey of new mortgage borrowers conducted by the Federal Housing Finance Agency (FHFA) and the Consumer Financial

²Fannie and Freddie then package many mortgages into mortgage-backed securities (MBS), which they sell to investors, such as pension funds and hedge funds.

³Refinancing does not require that the interest rate decrease. Some borrowers may choose to do a cash-out refinance even if interest rates are higher to access cash upfront. Similarly, some consumers might refinance at a higher fixed rate to extend the loan term, leading to a lower monthly payment.

⁴https://ycharts.com/indicators/mortgage_originations_refinancing_shares; accessed Jan 31, 2025

⁵<https://mortgage.ice.com/resources/data-reports/january-2021-mortgage-monitor>; accessed March 12, 2025

⁶The prevailing mortgage rate is typically only available for common loan configurations (e.g., 30-year or 15-year fixed-rate loans), assuming a strong credit profile (e.g., credit score, loan-to-value ratio). Since borrowers' credit profiles vary, they need to contact lenders to obtain personalized quotes.

Protection Bureau (CFPB),⁷ we find that among those who took out a conventional refinance loan during our sample period, 49.7% of consumers reported that they seriously considered only one lender before applying for a loan.⁸

We focus on refinance mortgages because consumers often rely on real estate agents or mortgage brokers to choose a lender for their first mortgage at the time of the home purchase, which may limit the impact of advertising. In contrast, refinancing decisions are typically made independently by consumers.

We use a publicly available dataset collected under the Home Mortgage Disclosure Act (HMDA) to obtain refinance mortgages that were originated between 2018 and 2021. HMDA is a U.S. federal law that requires mortgage lenders to disclose detailed information about their mortgage lending activities. The data collected under this law cover approximately 90% of total mortgage originations, with reporting exemptions for small lenders (Bhutta, Laufer, and Ringo, 2017). This dataset provides comprehensive loan-level information, including the originating lender, year of origination, loan size, and loan type. The data also include the borrower’s race, which allows us to study how different racial groups respond to minority representation in ads, as well as the property’s census tract, which we match with census tract-level voting outcomes to analyze heterogeneous effects by political leaning.

We provide a detailed overview of the data cleaning process used to obtain our sample of conforming refinance mortgages in Online Appendix A, where we outline the procedure and the number of loans remaining at each step. In addition to selecting conforming loans, we further restrict our sample to loans originated in the top 101 Designated Market Areas (DMAs) to align with the TV advertising data (discussed in Section 2.2). Moreover, to analyze heterogeneous effects across different racial groups, we exclude loans with missing or joint race information. Our final sample consists of 9.67 million refinance mortgage loans. Approximately 21.9% of loans were originated for minority borrowers, with Black, Hispanic, and Asian borrowers accounting for 4.4%, 8.1%, and 9.4% of the market, respectively.

To study the choice of lenders, we focus on the top 30 lenders, which together account for approximately 50% of all refinanced mortgage originations, and group the remaining smaller lenders into an “outside” option. Table 1 presents the list of lenders, ranked by market share, in Column

⁷<https://www.fhfa.gov/data/national-survey-mortgage-originations-nsmo-public-use-file>; Accessed March 13, 2025

⁸The exact question we used is, “How many different mortgage lenders/brokers did you seriously consider before choosing where to apply for this mortgage?”

Table 1: Market Share and TV Advertising Spending by Lender per Year

Mortgage lender	Market Share	TV Ad Spending (million \$)	Proportion of National Ads
	(1)	(2)	(3)
Rocket Mortgage	10.64%	200.8955	98.70%
United Wholesale Mortgage	5.56%	0.0443	0%
Wells Fargo	3.92%	6.8368	70.55%
JP Morgan Chase	3.64%	0.0005	0%
LoanDepot	2.71%	24.8666	99.80%
Nationstar	1.84%	1.4282	97.34%
Bank of America	1.67%	0	—
Caliber Home Loans	1.48%	0	—
US Bank	1.27%	0.3295	88.72%
Fairway Independent	1.26%	0.7522	0%
PennyMac	1.16%	0	—
Guaranteed Rate	1.11%	7.0856	97.28%
Flagstar Bank	1.09%	0.0371	0%
Home Point Financial	1.06%	0	—
Freedom Mortgage	0.98%	0.0067	0%
Newrez Mortgage	0.85%	0	—
Provident Funding	0.83%	0	—
AmeriSave Mortgage	0.76%	7.8441	94.98%
Better Mortgage	0.74%	0.0656	100%
Citizens Bank	0.72%	0.0104	0%
CrossCountry Mortgage	0.72%	0.2812	5.49%
PNC Bank	0.70%	0.0007	75.96%
Broker Solution	0.68%	0.0022	0%
Cardinal Financial	0.68%	0.0057	0%
Finance of America	0.67%	0.0157	0%
Guild Mortgage	0.59%	0.0409	0%
Fifth Third Bank	0.51%	0	—
Huntington Natl. Bank	0.48%	0	—
Movement Mortgage	0.48%	0.0011	0%
American Financing	0.41%	4.2005	0.74%
Total	49.21%	254.7511	95.79%

(1).⁹ For each borrower, we define their choice set based on location and time. If a lender originated at least one loan in a given DMA in a particular year, we assume it was available for all borrowers in that market. Since our analysis focuses on large lenders, they are generally available in most of the 101 DMAs, and the average borrower in our sample has access to 30.41 lenders, including the

⁹We exclude BB&T and SunTrust from the list, as they merged into Truist in December 2019. During the transition, they ran separate ad campaigns but reported to HMDA under the new name Truist, making it challenging to match the advertising data with the loan origination data.

outside option.

We find little change in product offerings within lenders over time. Specifically, we looked to see if any combinations of (1) cash-out vs. non-cash-out, (2) fixed- vs. adjustable-rate, and (3) different term lengths were observed in later years but not in earlier years at the lender level. These would be the maximum number of new loan products that a lender could have introduced in any year, although many zeros in our data may occur due to low demand (for example, there are loans with very unusual term lengths). We find that, at a maximum, the number of new loan combinations makes up less than 0.5% of loans in any year.

We define the variables in Columns (2) and (3) of Table 1 in the sections below.

2.2 Mortgage TV Advertising Data

We obtain TV advertising data from Vivvix (formerly Kantar Media) for 2018-2021, the same time as the mortgage data. This data includes monthly advertising spending at the lender-DMA-ad creative-channel level across the top 101 Designated Market Areas (DMAs). DMAs are geographic TV markets defined by Nielsen to measure TV ratings. Each DMA typically consists of multiple counties, with a major city at its center and surrounding smaller counties.

TV advertising is the primary marketing channel in this market, with the top lenders collectively spending \$255 million per year on TV ads. Column (2) of Table 1 reports TV ad spending by lender. There is substantial variation across lenders. Rocket Mortgage is the largest advertiser, followed by LoanDepot and Wells Fargo. In contrast, some lenders, such as JP Morgan Chase and Bank of America, allocate little or no budget to TV advertising.

Among the \$255 million spent annually on TV ad spending, \$244.1 million (or 95.8%) are national ads that air across all DMAs, and the remaining \$10.7 million (or 4.2%) are local ads that air in specific DMAs. The choice between national and local ads varies across lenders. Column (3) of Table 1 reports the proportion of national ads for each lender. Not surprisingly, lenders that operate nationally, such as Rocket, Wells Fargo, US Bank, and LoanDepot, tend to spend more heavily on national ads. In contrast, relatively smaller lenders, such as Fairway Independent, American Financing, and Movement, focus primarily on local ads.

To estimate the impact of TV ad spending on lender choice, we calculate each lender’s ad spending per capita within each DMA. For local ads, spending is divided by the DMA population (Shapiro 2018; Tsai and Honka 2021), and for national ads, spending is divided by the national

population. A lender’s total ad spending per capita in a given DMA is the sum of its national and local ad spending per capita within that DMA. When merging with the loan origination data, we shift the ad spending data back by one month to reflect the time lag between loan application and origination. Across all lenders and DMAs, the average annual ad spending per capita is \$0.0297.

In addition to the ad spending data, we also collect the video files of the TV ad creatives. The top 30 lenders aired about 1440 unique ad creatives during our sample period. The average time that an ad creative airs is 4.6 months and the median is 3 months. We utilize these ad videos to determine whether and to what extent they feature minority actors. In addition to race, we extract visual and textual features from these video ads, as detailed in Section 3.

Note that the focus of this paper is on TV advertising. We focus on this and not on other activities, such as online advertising, for a few reasons. First, we do not observe the ad creatives for online advertising, so it is hard to use the online advertising data to get insights into how race within an advertisement affects sales. Second, TV advertising is the predominant advertising channel in this market. We verify this by collecting data on each lender’s social media ads (from Meta, available starting in May 2018) and online display ads (from Vivvix). Among the top 30 lenders, the level of ad spending on Meta and online display ads is about 0.015%, and 9.9% of the spending on TV advertising, respectively, although some firms do focus more on online advertising.¹⁰ Further, 90% of online ads are classified as being on national websites, while 10% are classified as being on local websites. Thus, online spending is relatively small overall, and most of this spending is at the national level, which will be captured by the firm-year fixed effects in our model.

2.3 Race Detection and Minority Representation Measure

We leverage pre-trained computer vision algorithms from Clarifai Inc. to identify the race of actors in video ads. For each video, we extract one frame per second to generate frame-level images. The Clarifai algorithm detects faces, crops an image for each detected face, and assigns probabilities that the face belongs to each racial group. Figure 1 shows two examples. In the first image, Clarifai detects two faces, predicting the male actor is Black with 86% probability and the female actor is Black with 99% probability. In the second image, a single face is detected, with the actor predicted to be East Asian with 80% probability. We have manually checked a number of predictions and found Clarifai’s algorithms to be highly accurate.

¹⁰Detailed ad spending data by lender is provided in Online Appendix B.

Figure 1: Clarifai Examples



Using the race predictions from Clarifai, we align racial categories with those in the HMDA data. First, we assign each actor as belonging to the race with the highest probability given by Clarifai. To minimize measurement errors, we exclude cases where the highest predicted probability is below 70%, similar to previous studies (e.g., [An and Kwak, 2019](#); [Guitart and Stremersch, 2021](#)). Then we merge “White” and “Middle Eastern” into the White category and group “East Asian”, “Southeast Asian”, and “Indian” into the Asian category. This results in four groups: White, Black, Hispanic, and Asian, which match the racial categories that are given in the HMDA data.¹¹

Using this frame-level information, we compute racial representation in each video using both the duration each race appears on screen and the extent of screen sharing when multiple actors are present. Let $f = 1, \dots, F$ denote frames containing human faces, and let J^f represent the number of actors in frame f . The race of the j^{th} actor in frame f is denoted as $R_j^f \in \{W, B, H, A\}$,

¹¹In our study, we use “race” to refer to both race and ethnicity, as classifications are based on actors’ appearances. Thus, we treat Hispanic as a racial group.

where W =White, B =Black, H =Hispanic, and A =Asian. When a frame contains multiple actors ($J^f \geq 2$), we divide the screen share equally and assign each actor a proportion $\frac{1}{J^f}$. The racial representation measure for each video is then calculated as:

$$Share^{Race} = \frac{1}{F} \sum_{f=1}^F \left(\frac{1}{J^f} \sum_{j=1}^{J^f} \mathbb{1}\{R_j^f = Race\} \right), \quad (1)$$

where $Race \in \{W, B, H, A\}$ represents each racial group, and $\mathbb{1}\{R_j^f = Race\}$ is an indicator function that equals 1 if the j^{th} actor in frame f belongs to the specific $Race$, and 0 otherwise.

Consider a 30-second ad as an example. Suppose a White actor appears alone in the first 14 seconds, followed by two actors, one White and one Black, sharing the screen for the next 14 seconds, with the lender's logo in the final 2 seconds. In this example, the total number of frames with human faces is $F = 28$. The share of White actors is calculated as $Share^W = \frac{1}{28} (14 \cdot 1 + 14 \cdot 0.5) = 0.75$ because the first 14 seconds feature only a White actor, and the next 14 seconds feature both a White and a Black actor, each receiving an equal screen share of $\frac{1}{2}$. Similarly, the share of Black actors is $Share^B = \frac{1}{28} (14 \cdot 0 + 14 \cdot 0.5) = 0.25$ because the Black actor appears in the latter 14 seconds and shares the screen with the White actor.

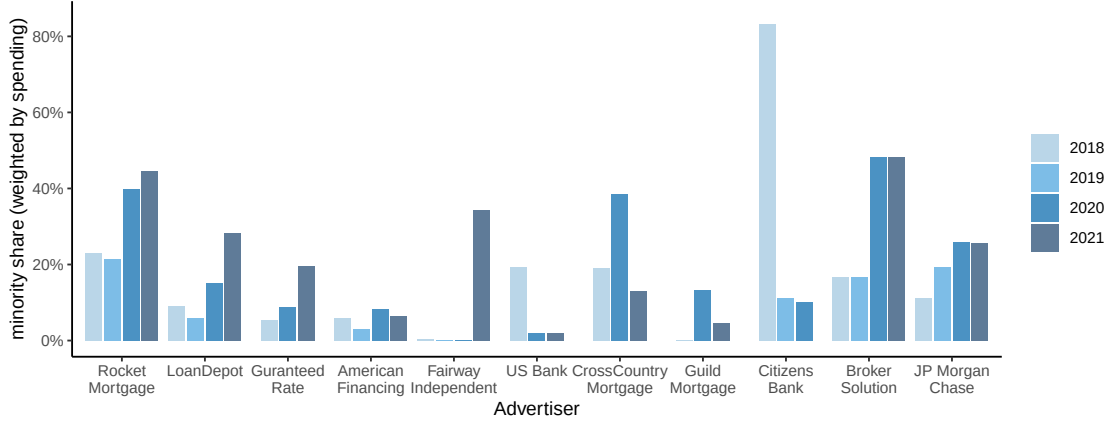
Across 1,444 unique video ads in our data, White actors account for approximately 79.7% of total screen time in TV ads, Black actors account for 13.0%, Asian actors account for 6.4%, and Hispanic actors account for 0.9%.¹² In our analysis, we aggregate the shares of Black, Hispanic, and Asian representation to measure the total minority representation in each ad.

Over the four years in our sample, we observe that the average minority share, weighted by ad spending, increases from 23% in 2018 and 21% in 2019 to 35% in both 2020 and 2021. A similar trend of increasing minority representation is observed in online display advertising (Hartmann, Netzer, and Zalta, 2023). Figure 2 plots the weighted average of minority share, with ad spending as the weight, for lenders that advertise in at least three out of the four years. While we observe an overall trend of a higher minority share in 2020 across multiple lenders, there is still heterogeneity among them.

In addition to time trends, Figure 2 also reveals differences in minority share in ads across lenders. One might hypothesize that lenders adjust minority representation in ads based on the proportion of minority borrowers. Figure 3 explores this possibility by plotting the average minority

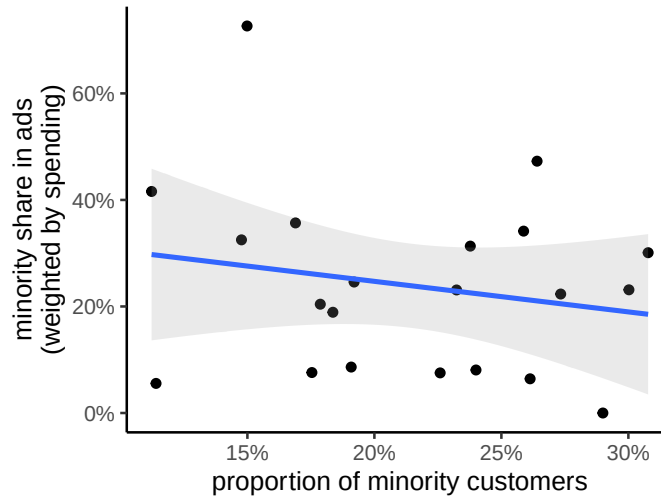
¹²The lower Hispanic representation is partly due to limitations in Clarifai's race detection algorithm, which primarily identifies individuals with more indigenous Hispanic features. As a result, many Hispanic individuals may be classified as White. A similar issue is discussed in Davis et al. (2019).

Figure 2: Minority Share in Ads by Lender Over Time



share in ads (weighted by spending) against the proportion of minority customers by lender. While both metrics show substantial variations across lenders, we do not observe a clear relationship between the two. Therefore, although minority share in ads varies across lenders, this variation does not appear to be driven by the composition of their customer base.

Figure 3: Minority Share in Ads and Proportion of Minority Borrowers by Lender



2.4 Estimation Sample

For estimation, we aggregate advertising spending and minority share variables at the lender-DMA-year level to align with the HMDA data. Ad spending is simply the sum of spending per capita across different ads, including both national and local ones. Minority share is computed as the weighted average of minority share across ads, with ad spending in the DMA as the weight.

Table 2 presents summary statistics of the key variables in the estimation sample. The average ad spending per capita among the top 30 lenders is \$0.0297. On average, minority actors occupy 15.9% of screen time in mortgage ads, which is slightly lower than the proportion of minority borrowers in this market. Our sample includes 9,668,300 loans, 21.9% of which are taken out by minority borrowers. Since we do not observe individual-level political ideology, we proxy it using the census tract-level voting outcomes based on the property’s location (Amos, McDonald, and Watkins, 2017; Bryan, 2022). We define Dem_i as the difference in vote shares between Biden and Trump in the 2020 presidential election: $Dem = (\text{Biden votes} - \text{Trump votes}) / (\text{Biden votes} + \text{Trump votes})$. Votes for third-party candidates are excluded from this calculation. Thus, Dem must lie within the range from -1 to 1, with higher values indicating that the consumer lives in a more liberal-leaning community.

Table 2: Summary Statistics of the Estimation Sample

	Mean	Min	25%	Median	75%	Max
<i>At borrower-lender level</i>						
Ad spending (\$ per capita)	0.0297	0	0	0	0	10.9793
Minority share in ads	0.1594	0	0.0206	0.1438	0.2664	1
<i>At borrower level</i>						
Minority borrowers	0.2187	0	0	0	0	1
Dem vote share	0.0740	-0.9744	-0.1854	0.0630	0.3247	0.9706

Notes: Minority share in ads is conditional on positive ad spending.

3 Other Video Ad Features

While our main goal is to estimate the impact of minority representation in advertising on consumers’ choice of lenders, the presence of different races in ads may be correlated with other confounding factors, such as visual elements or advertising messages. These potential differences could introduce omitted variable bias if not properly addressed. We seek to account for the possibility of such correlation by controlling for a large number of video features. In this section, we

describe how we extract the high-dimensional visual and textual features from ads.

3.1 Visual Features

We start by pre-processing the videos before extracting visual features from them. A typical video in our data has 1.37 billion pixel values (960 pixels for width $\times$ 540 for height $\times$ 30 seconds $\times$ 30 frames per second $\times$ 3 color channels).¹³ To reduce the computational burden, we sample one frame every 5 seconds, which results in an average of 6 frames per video since most video ads are 30 seconds long. Following standard practices in computer vision, we resize the frame-level data to have 150,528 pixel values per frame (224p for width $\times$ 224p for height $\times$ 3 color channels).

There are two broad approaches to extracting features from images. The first approach is to extract a number of researcher-defined features, which are often interpretable. For example, [Zhang et al. \(2022\)](#) examine how 12 image features, such as brightness and situation, impact the demand for Airbnb properties. This approach can offer interpretable insights into what is captured in the image data. However, it is often not straightforward to determine which features to extract. Furthermore, unless pre-trained algorithms are available, researchers would need to train their own models to extract desired features, which are more likely to be subject to measurement errors.

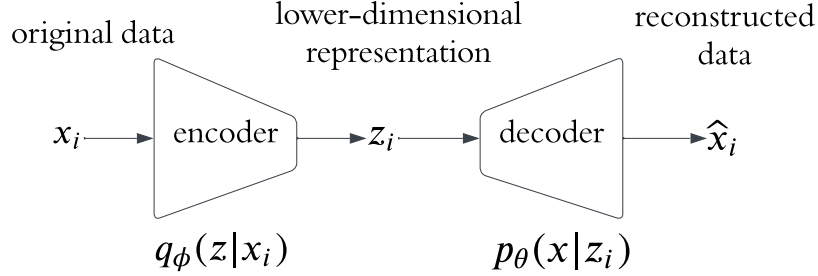
The second approach is to obtain embeddings that represent image data. These embeddings are numeric vectors that are not easily interpretable. However, unlike the first approach, researchers do not need to manually define a list of features to extract. Instead, it relies on a data-driven approach to capture as much relevant information as possible. Since our primary goal is to use these features as control variables to mitigate potential confounding factors, we prioritize comprehensiveness over interpretability. Therefore, we use the second approach for extracting image features.

We use a variational autoencoder (VAE) to generate a lower-dimensional representation of the images ([Kingma and Welling, 2013](#); [Rezende, Mohamed, and Wierstra, 2014](#)). The structure of a VAE is illustrated in Figure 4. An image x_i , represented by high-dimensional pixel-level values ($x_i \in \mathbb{R}^d$), passes through an encoder network $q_\phi(z|x_i)$, which generates a lower-dimensional latent representation z_i . More specifically, the encoder network generates a stochastic representation by outputting the parameters (mean and variance) of the distribution $q_\phi(z|x_i)$, which is a Gaussian probability density. $z_i \in \mathbb{R}^k$ is then sampled from this distribution and has a significantly smaller dimension compared to x_i . The low-dimensional representation z_i is then fed into the decoder

¹³The vast majority of the video ads have a resolution of 960p $\times$ 540p with a frame rate of 30 frames per second.

network $p_{\theta}(x|z_i)$, which generates the reconstructed image $\hat{x}_i \in R^d$. The parameters ϕ and θ correspond to the encoder and decoder networks, respectively.

Figure 4: Illustration of Variational Autoencoder



Intuitively, a VAE is a semi-supervised machine learning model that aims to reconstruct the original image using a low-dimensional latent representation to minimize the reconstruction error. More formally, a VAE is trained by minimizing the following loss function:

$$L_{\phi,\theta}(x) = -E_{z \sim q_{\phi}(z|x)} [\log(p_{\theta}(x|z))] + D_{KL}(q_{\phi}(z|x) || p(z)), \quad (2)$$

where the first term is the expected log-likelihood of the data x given the latent representation z . The expectation is taken over the encoder’s distribution over the representation z , so that it depends on both the encoder parameters ϕ and the decoder parameters θ . This term is known as the reconstruction loss since it encourages the decoder to reconstruct the original data by maximizing the likelihood. The second term is the regularization loss that ensures that latent representation follows a smooth distribution. This is the Kullback–Leibler divergence between the encoder’s distribution $q_{\phi}(z|x)$ and the prior standard normal distribution $p(z)$. We refer interested readers to [Kingma, Welling et al. \(2019\)](#) for further details on the model and estimation.

We implement a VAE using the frame-level video data. We train a convolutional encoder-decoder neural network that includes fully connected layers in between to generate latent representation z . We choose a 100-dimensional latent representation for the images in our data, as increasing the dimensionality further provides minimal improvement in the loss function. We provide detailed information about the network structure, training process, and loss function values in [Online Appendix C](#).

After training the model, we use the trained encoder to generate a lower-dimensional represen-

tation z for each frame. We then aggregate these z -values at the video level by taking the average across all frames within the video. To align with our mortgage origination data, we further aggregate these video-level image embeddings to the lender-DMA-year level by calculating the weighted average across both national and local ads in the DMA, with ad spending per capita as the weight.

3.2 Textual Features

To extract features that capture the messages conveyed in the ads, we first obtain the video transcripts using the Amazon Transcribe API, a speech recognition service that converts the audio content into text. Similar to image analysis, one could seek to identify interpretable features or obtain text embeddings that are not directly interpretable but contain comprehensive information. Since the textual features will serve as control variables, we choose the second approach by using a pre-trained large language model to obtain text embeddings.

To represent the textual content of the ads, we utilize a pre-trained embedding model, “text-embedding-ada-002,” from OpenAI.¹⁴ The ada-002 model generates 1,536-dimensional embeddings per document. Since this dimensionality is still relatively large, we compress the embeddings into 100 dimensions using another VAE model. Increasing the dimensionality beyond 100 shows only relatively small improvements in the loss function. Further details can be found in C. As with image embeddings, we aggregate these text embeddings to the lender-DMA-year level by taking the weighted average across both national and local ads in the DMA, with the ad spending per capita as the weight.

Although we prefer the embeddings approach for comprehensiveness, we also explore how ad content varies for ads with greater minority representation. To do so, we employ zero-shot learning, which allows content classification based on pre-specified categories. By manually reviewing ad samples, we identify three common topics: 1) rate/savings, such as “we can help you take advantage of today’s low mortgage rates and save money,” 2) ease of application, such as “with our tools the entire process was quick and simple,” and 3) brand-focused messages, such as “ranked #1 in the US for consumer satisfaction by J.D. Power.”¹⁵ Overall, the (spending weighted) proportions of the

¹⁴While OpenAI offers multiple embedding models, such as “davinci”, “curie”, and “babbage” that are better suited for different tasks (e.g., clustering or search), they recommend “ada-002” for most use cases. For more information, see <https://platform.openai.com/docs/guides/embeddings>. Accessed July 13, 2023

¹⁵We use the following keywords to specify the three categories: 1) low mortgage rate, interest saving, lower payments, save on mortgage, 2) online application, digital mortgage, quick approval, easy and simple process, and 3) trusted lender, customer satisfaction, expert advice, community-focused, professional. For implementation, we use a pre-trained model from Hugging Face, available at

three ad topics are relatively balanced at 36%, 27%, and 37%, respectively. The rate/savings topic is positively correlated with minority representation in ads (0.15, weighted by ad spending), while the ease of application and brand-focused messages have negative correlations of -0.06 and -0.09, respectively, with minority representation. The correlations of the messages with the minority representation in ads demonstrate the importance of controlling for ad content in our analysis.

4 Model and Estimation Strategy

In this section, we present the model and estimation strategy in order to understand how consumers respond to minority representation in ads. In Section 4.1, we describe the model to estimate the impact of minority share in ads on consumers' choice of lenders. The model incorporates lender-DMA and lender-time fixed effects, as well as visual and textual features detailed in Section 3. In Section 4.2, we describe the double machine learning (Double ML) estimator to obtain consistent estimates for the key parameters of interest while effectively handling the high-dimensional control variables.

4.1 Empirical Model

We start by outlining the model to estimate the impact of ad spending and minority representation in ads on consumer choices. Equation (3) describes the main model:

$$y_{i,j} = \beta_1 \cdot Ad_{j,m_i,t_i} + \beta_2 \cdot Ad_{j,m_i,t_i} \cdot MS_{j,m_i,t_i} + \delta_{j,m_i} + T_{j,t_i} + g(X_{j,m_i,t_i}) + e_{i,j}, \quad (3)$$

where $y_{i,j}$ is an indicator variable that equals 1 if consumer i chooses lender j and 0 otherwise. Ad_{j,m_i,t_i} denotes ad spending per capita by lender j in the DMA where consumer i resides (m_i) and the year the consumer obtains a refinance loan (t_i), while MS_{j,m_i,t_i} represents the corresponding minority share in the ads, as defined in Section 2.3.¹⁶ X_{j,m_i,t_i} captures the visual and textual features of the ads, as described in Section 3, and $g(X_{j,m_i,t_i})$ represents a flexible function of these nuisance variables. $e_{i,j}$ denotes an idiosyncratic error term.

We include lender-DMA fixed effects δ_{j,m_i} and lender-year fixed effects T_{j,t_i} in the model. The lender-DMA fixed effects account for local, time-invariant confounding factors that might be correlated with advertising choices. For example, some lenders may have historically given more

<https://huggingface.co/MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli>. Accessed Feb. 6, 2025

¹⁶We conduct robustness checks using alternative definitions of minority share. See Online Appendix D.

loans, which may affect which lenders are top of mind for consumers in an area. Similarly, some lenders may have higher demand because the lender has more offices in a market or a longer history in a particular market. The lender-year fixed effects capture global, time-varying confounding factors, such as lenders choosing to advertise more, or including more minorities in ads during certain time periods, such as the increase in minority representation in advertising that occurred after the murder of George Floyd.

We use a linear specification for ad spending because of the presence of zero advertising levels in the data. In the past, many studies would use a $\log(1 + Ad)$ functional form for advertising. However, it has recently been shown that doing so causes measurement issues where the magnitude is sensitive to the scaling of the variables (Chen and Roth, 2024). We also tested the implications of scaling advertising and found that the elasticities varied with the advertising scaling that we used. Thus, we follow several recent papers in the advertising literature and use a linear functional form for advertising (e.g., Hristakeva, 2023; Knight, Hsieh, and Bart, 2025).

The key parameters of interest are β_1 and β_2 . β_1 measures the baseline level of advertising effectiveness when the minority share is zero, and β_2 captures how advertising effectiveness varies with different levels of minority share in advertisements. If the nuisance features, X_{j,m_i,t_i} , were excluded, the model would reduce to a standard fixed effects specification for estimating advertising effectiveness. The advertising effects are thus identified from the time-varying advertising by lenders within each market, as well as the within-time variation across markets. Following the prior literature on advertising, we show in Online Appendix E that there is sufficient residual variation in advertising to allow for the estimation of these parameters. To interpret the advertising effects as causal, we are essentially assuming that the remaining variation in ads is random.

In addition to the main effects, we analyze how the results vary based on consumers' race and political ideology. The precise functional form we use is given in equation (4). $\mathbb{1}\{M_i\}$ denotes an indicator variable that equals 1 if consumer i belongs to a racial minority group and 0 otherwise (i.e., non-Hispanic White). Dem_i represents the difference in vote shares between Biden and Trump, as defined in Section 2.3. The other variables are defined in the same way as in the main model in Equation (3).

$$\begin{aligned}
y_{i,j} = & (\beta_1 + \beta_2 \cdot \mathbb{1}\{M_i\} + \beta_3 \cdot Dem_i) \cdot Ad_{j,m_i,t_i} \\
& + (\beta_4 + \beta_5 \cdot \mathbb{1}\{M_i\} + \beta_6 \cdot Dem_i) \cdot Ad_{j,m_i,t_i} \cdot MS_{j,m_i,t_i} \\
& + g(X_{j,m_i,t_i}) + \delta_{j,m_i} + T_{j,t_i} + e_{i,j}.
\end{aligned} \tag{4}$$

4.2 Estimation Strategy

We use the Double ML estimator to estimate the β 's while accounting for high-dimensional covariates $g(X_{j,m_i,t_i})$. Double machine learning has become increasingly popular in economics and marketing for causal inference using observational data with high-dimensional control variables (e.g., [Dube et al., 2020](#); [Ellickson, Kar, and Reeder III, 2022](#); [Gordon, Moakler, and Zettelmeyer, 2022](#); [Gershon and Jiang, 2024](#)). In this section, we start by briefly discussing the key idea behind the Double ML estimator and then describe how we use it to estimate the model with both fixed effects and visual and textual features in our application.

A naive approach to incorporating $g(X_{j,m_i,t_i})$ is to estimate the flexible function $\widehat{g(\cdot)}$ using a machine learning model and plug it into the regression. However, this approach introduces regularization bias in the estimates of $\widehat{\beta}$ due to regularization in machine learning, which results in $\mathbb{E}(\widehat{g(\cdot)}) \neq g(\cdot)$. This bias diminishes with larger sample sizes but at a slower rate than $n^{-1/2}$. Furthermore, the flexible nature of machine learning models can lead to overfitting, introducing further bias.

The Double ML estimator addresses regularization bias and overfitting bias through double orthogonalization and sample-splitting. We describe the process in more detail in the ensuing paragraphs, but the high-level idea is to leverage machine learning models to “partial out” the effects of high-dimensional control variables from both the outcome and treatment variables. Such a process ensures robustness against first-order errors in machine learning predictions. Additionally, sample-splitting prevents overfitting by training models on one part of the data and estimating residuals using another.

We now discuss our implementation of the Double ML estimator to estimate the β 's in Equations (3) and (4). We use a linear probability model as an approximation for a multinomial choice model, both because estimating a multinomial choice model becomes quantitatively difficult with high-dimensional covariates and a large number of fixed effects, and because some of the fixed effects are only associated with a limited number of observed purchases, which causes bias in a non-linear function such as a multinomial logit, but not in a linear probability model (e.g., [Neyman and Scott, 1948](#); [Hahn and Newey, 2004](#)).¹⁷

We start by describing the orthogonalization procedure. First, we run the regression shown in

¹⁷The predicted values for y using the linear probability model range from -0.024 to 0.912, with only 3.1% of observations falling outside the 0-1 interval, suggesting that the linear probability model is a reasonable choice.

Equation (5) to partial out the effects of all the nuisance components from the dependent variable:

$$y_{i,j} = g(X_{j,m_i,t_i}) + \delta_{j,m_i} + T_{j,t_i} + \epsilon_{i,j}. \quad (5)$$

These nuisance variables include the lender-market fixed effects δ_{j,m_i} , the lender-time fixed effects T_{j,t_i} , and a flexible form of the visual and textual features $X_{i,j}$. We implement this using a neural network with three branches, which we will describe in more detail below. Using the estimated neural network model, we obtain the residual $\tilde{y}_{i,j} = y_{i,j} - \mathbb{E}(y|X_{j,m_i,t_i}, \delta_{j,m_i}, T_{j,t_i})$, which represents the residual variation of y after accounting for these nuisance components.

Similarly, we estimate a separate neural network to predict each low-dimensional treatment variable of interest using the same model structure, as shown in Equation (6):

$$D_{j,m_i,t_i} = m(X_{j,m_i,t_i}) + \delta_{j,m_i} + T_{j,t_i} + \epsilon_{j,m_i,t_i}. \quad (6)$$

In our setting, D_{j,m_i,t_i} represents Ad_{j,m_i,t_i} and $Ad_{j,m_i,t_i} \cdot MS_{j,m_i,t_i}$. After accounting for lender-market fixed effects δ_{j,m_i} , lender-time fixed effects T_{j,t_i} , and visual and textual features X_{j,m_i,t_i} , we compute the residual variation as $\tilde{D}_{j,m_i,t_i} = D_{j,m_i,t_i} - \mathbb{E}(D|X_{j,m_i,t_i}, \delta_{j,m_i}, T_{j,t_i})$. A separate model is trained for each low-dimensional treatment variable. For example, to estimate Equation (3), we train one model to predict Ad_{j,m_i,t_i} and a different model to estimate $Ad_{j,m_i,t_i} \cdot MS_{j,m_i,t_i}$, resulting in two residual variables $\tilde{\mathbf{D}} = [\tilde{Ad}, \tilde{Ad} \cdot \tilde{MS}]$.

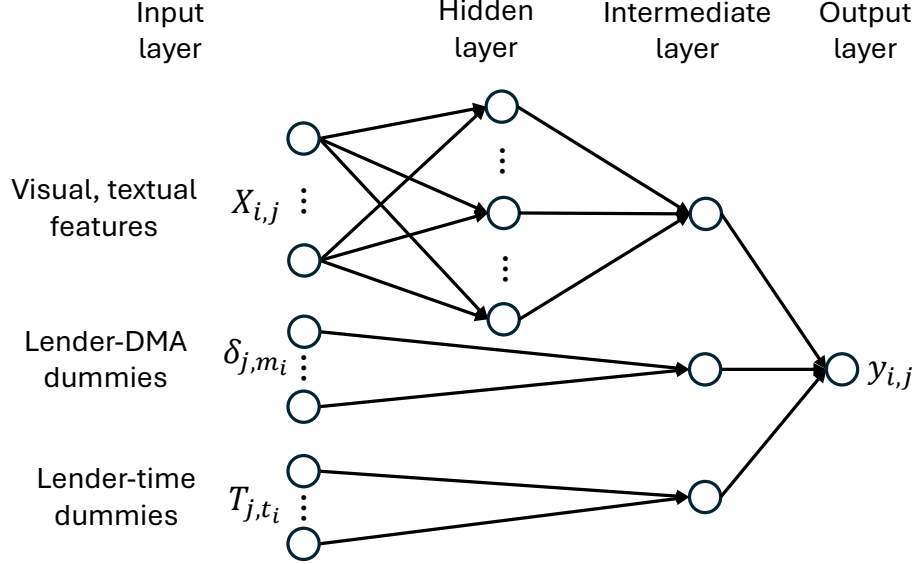
The residuals $\tilde{y}_{i,j}$ and $\tilde{\mathbf{D}}_{j,m_i,t_i}$ capture the variations in the outcome and treatment variables, respectively, after “partialling-out” the fixed effects and high-dimensional features. The Double ML estimate is then obtained by the regression in Equation (7):

$$\tilde{y}_{i,j} = \beta \cdot \tilde{\mathbf{D}}_{j,m_i,t_i} + \eta_{i,j}. \quad (7)$$

The estimated β ’s from this equation remain robust to imperfect machine learning predictions (e.g., regularization bias).

The procedure outlined above addresses regularization bias, but the estimates of $\hat{\beta}$ may still be biased due to overfitting. This issue is solved by sample-splitting, where we partition the data into $K = 3$ folds. For each fold, k , we train a neural network model based on Equations (5) and (6) using data from the two other folds (i.e., excluding fold k). The fitted models are then used to estimate the residuals, and we use Equation (7) to obtain β^k for the k -th fold. This ensures that the data used to estimate the residuals is separate from the data used to obtain the causal effects. Finally, the Double ML estimate is obtained by averaging the K fold-specific estimates: $\hat{\theta} = \frac{1}{K} \sum_k \hat{\theta}^k$. See

Figure 5: Illustration of the Neural Network Structure



[Chernozhukov et al. \(2018\)](#) for more details about the Double ML process.

We now describe the neural network structure used in our model estimation. As shown in Figure 5, the neural network consists of three separate branches. One branch contains the lender-market fixed effects δ_{j,m_i} , another branch contains the lender-time fixed effects T_{j,t_i} , and the last branch contains the visual and textual features X_{j,m_i,t_i} . This architecture aligns with the model specification in Equations (5) and (6). The outputs from the three branches in the intermediate layer are concatenated to form the final prediction. We use the same network structure to predict the outcome variable as well as each of the treatment variables. By maintaining separate branches, the model is restricted from learning undesirable interactions, such as lender-market-time effects, which would fully absorb the advertising variation. A similar network structure is proposed by [Cheng et al. \(2016\)](#) for recommender systems.

Lastly, the standard approach of calculating standard errors with the Double ML estimator, which assumes independent and identically distributed (i.i.d.) errors, is not applicable in our case. This is because the observations in our dataset are correlated, as each consumer chooses a single lender for refinancing. To properly account for the correlation structure, we calculate the standard errors by bootstrapping the data at the loan level ([Cameron and Miller, 2015](#)). More specifically, we create 30 bootstrapped datasets by resampling loans with replacement, estimate coefficients for each of these 30 resampled datasets, and then compute the standard deviation of the bootstrap

estimates, which is the bootstrap standard error of the coefficients.

5 Empirical Results

5.1 Main Results

Table 3 presents the results from our main model. While the baseline effect of ad spending on lender choice (β_1) is positive but not statistically significant, the interaction between ad spending and minority share (β_2) is positive and statistically significant. This result indicates that greater minority representation increases the overall effectiveness of advertising.

Table 3: Impact of Minority Representation in Ads

	Lender choice
$\beta_1 : Ad$	0.00594 (0.01539)
$\beta_2 : Ad \cdot MS$	0.07968*** (0.02821)
Num. of Unique Loans	9,668,300

Notes: MS denotes the minority share in ads.

Standard errors are clustered at the loan level.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To interpret the effect size of the estimates, we calculate the implied advertising elasticity of demand with respect to ad spending, which is a commonly used measure of advertising effectiveness. With simple algebra, the advertising elasticity can be derived as: $\frac{\partial y}{\partial Ad} \frac{Ad}{y} = \beta_1 \cdot \frac{Ad}{y} + \beta_2 \cdot MS \cdot \frac{Ad}{y}$. The first term represents the elasticity of ad spending in the absence of any minority representation, while the second term represents the incremental effect of the minority share in ads on ad elasticity. We calculate the average advertising elasticity of demand using the sample averages for all variables: $\overline{Ad} = 0.0297$, $\overline{MS} = 0.1595$, and $\overline{y} = 0.0329$, which yields an average advertising elasticity of 0.0168. This effect size is broadly in line with previous studies: the average elasticity is 0.023 for 288 consumer packaged goods (Shapiro, Hitsch, and Tuchman, 2021), 0.026 for cigarette product placement on TV (Goli et al., 2022), 0.030 for auto insurance (Tsai and Honka, 2021), 0.031 for antidepressant (Shapiro, 2022), 0.05–0.06 for satellite TV operators (Yang, Lee, and Chintagunta, 2021), and 0.08 for e-cigarettes (Tuchman, 2019).

We examine the impact of minority share on advertising effectiveness by comparing the average advertising elasticities at two levels of minority share: 15% (close to the median minority share at

the lender-DMA level) and 30% (about one standard deviation increase), while keeping ad spending at its sample average. As the minority share increases from 15% to 30%, the estimated elasticity increases from 0.0161 (s.e. = 0.0114) to 0.0269 (s.e. = 0.0097). This increase of 0.0108 (s.e. = 0.0038) is statistically significant from zero. This result suggests that increasing the minority share in advertising can increase the effectiveness of advertising at an economically meaningful level.

Next, we examine how the effects of minority representation in ads may vary by the borrower’s race and political ideology. Recall that the borrower’s race is directly identified in the data, but the political ideology is inferred from the borrower’s census tract. The results are presented in Table 4. The advertising effectiveness when there is no minority representation in ads is lower for minorities (β_2) and liberal-leaning consumers (β_3).¹⁸ Minority representation in ads has a larger impact on minority borrowers than on White borrowers (β_5), as well as a larger impact on liberal-leaning borrowers compared to conservative-leaning borrowers (β_6).

Table 4: Heterogeneous Effects based on Race and Political Leaning

	Lender choice
$\beta_1 : Ad$	0.01163 (0.01522)
$\beta_2 : Ad \cdot \mathbb{1}\{M\}$	−0.00391*** (0.00132)
$\beta_3 : Ad \cdot Dem$	−0.02973*** (0.00109)
$\beta_4 : Ad \cdot MS$	0.06650** (0.02766)
$\beta_5 : Ad \cdot MS \cdot \mathbb{1}\{M\}$	0.08114*** (0.00352)
$\beta_6 : Ad \cdot MS \cdot Dem$	0.02443*** (0.00308)
Num. of Unique Loans	9,668,300

Notes: MS denotes the minority share in ads; $\mathbb{1}\{M\} = 1$ for minority borrowers; Dem is defined by the census tract vote share. Standard errors are clustered at the loan level.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To help put the coefficients into perspective, we calculate how advertising elasticities with respect to ad spending change when the minority share in ads increases from 15% to 30% across different consumer groups. The calculation of group-specific advertising elasticities is based on the sample averages from the respective group.¹⁹ The results are presented in Table 5. Panel A

¹⁸The 25th and 75th percentiles of Dem are -18% and 32%, respectively.

¹⁹Among minority consumers, $\overline{Dem} = 0.255$; among White consumers $\overline{Dem} = 0.024$. Among liberal-

reports the results by race. When the minority share increases from 15% to 30%, the increase in advertising elasticity is higher among minority than White borrowers. Panel B groups consumers by their political leaning (whether $Dem > 0$). The increase in advertising elasticity is slightly higher among liberal-leaning than conservative-leaning consumers. Panel C further explores the interactions between consumers' race and political ideology. The increase in advertising elasticity is higher for minority consumers, with minority liberals responding only a little bit more than minority conservatives. White consumers have a smaller boost from minority representation in ads, with White liberals getting a slightly larger increase than White conservatives.

Table 5: Advertising Elasticity by Consumers' Race and Political Leaning

Minority share	Advertising elasticity	Difference	Advertising elasticity	Difference
Panel A: By Consumers' Race				
	<u>Minority consumers</u>		<u>White consumers</u>	
15%	0.0210 (0.0115)		0.0188 (0.0113)	
30%	0.0418 (0.0098)	0.0208 (0.0038)	0.0279 (0.0097)	0.0091 (0.0037)
Panel B: By Consumers' Political Leaning (census tract level)				
	<u>Liberal-leaning</u>		<u>Conservative-leaning</u>	
15%	0.0142 (0.0115)		0.0260 (0.0111)	
30%	0.0277 (0.0098)	0.0135 (0.0038)	0.0352 (0.0095)	0.0093 (0.0037)
Panel C: By Consumers' Race & Political Leaning (census tract level)				
	<u>Minority Liberal</u>		<u>Minority Conservative</u>	
15%	0.0181 (0.0115)		0.0314 (0.0114)	
30%	0.0393 (0.0098)	0.0212 (0.0038)	0.0509 (0.0098)	0.0195 (0.0038)
	<u>White Liberal</u>		<u>White Conservative</u>	
15%	0.0125 (0.0115)		0.0253 (0.0110)	
30%	0.0227 (0.0098)	0.0102 (0.0038)	0.0333 (0.0095)	0.0080 (0.0037)

Notes: The calculation of ad elasticities uses group-specific sample means. Standard errors (in brackets) are calculated using the bootstrap estimates.

5.2 Potential Concerns on Causal Interpretation

In this section, we discuss several potential concerns about the causal interpretation of the observational results and conduct additional analyses that mitigate these concerns.

leaning consumers ($Dem_i > 0$), $\overline{1\{M\}} = 0.304$, $\overline{Dem} = 0.326$; among conservative-leaning consumers ($Dem_i < 0$), $\overline{1\{M\}} = 0.109$, $\overline{Dem} = -0.255$. Among minority liberal consumers, $\overline{Dem} = 0.375$; among minority conservative consumers, $\overline{Dem} = -0.182$; among White liberal consumers, $\overline{Dem} = 0.304$; among White conservative consumers, $\overline{Dem} = -0.264$. Additionally, the group-specific values for $\bar{y}$ and $\bar{Ad}$ are as follows: minority: 0.0295, 0.0328; White: 0.0297, 0.0329; liberal: 0.0299, 0.0328; conservative: 0.0293, 0.0329; liberal minority: 0.0295, 0.0327; conservative minority: 0.0298, 0.0328; liberal White: 0.0301, 0.0329; conservative White: 0.0293, 0.0329.

5.2.1 Targeting of minority representation on local demographics

One concern some readers may have is that lenders could perhaps target ads featuring minority actors in DMAs with larger minority populations or in areas with more liberal-leaning consumers. Such targeting could affect our interpretation of the effect of more minority representation in advertising since it would be possible that the coefficient we observe is really the result of minority or liberal consumers getting more exposure rather than the effect of the content.

We test whether there is evidence of such targeting by aggregating the minority share in ads at the DMA-year level and regressing it on the minority population share and the political leaning of the market. The results are presented in Table 6. Column (1) includes all ads (both national and local), while Column (2) focuses only on local ads. The coefficients for minority population share and political leaning are not statistically significant in either column. In terms of effect sizes, with minority population share going from 20% to 45% (about 25% to 75% percentiles) would increase the minority share in ads by only 0.3% overall, or by 1.5% for local ads. The magnitude of the *Dem* variable is also small and in the opposite direction that we are worried about.

We further examine evidence of dynamic targeting following the murder of George Floyd by interacting the key variables with pre-Floyd and post-Floyd dummies. The results are presented in Columns (3) and (4) of Table 6. In Column (3), which includes both national and local ads, the post-Floyd dummy is larger, consistent with an increase in minority share in ads post-Floyd. The interaction terms with minority population share and *Dem* are insignificant and small. When focusing on local ads in Column (4), there is no longer an increase in minority share post-Floyd. The interaction terms are not significant but somewhat larger in magnitude. To understand the effect size in Column (4), with *Dem* going from -20% to 15% (about 25% to 75% percentiles), the minority share in ads would decrease by 3.5% pre-Floyd and increase by 2.5% post-Floyd. Similarly, with minority population share going from 20% to 45% (about 25% to 75% percentiles), the minority share in ads would increase by 1.2% pre-Floyd and by 2.0% post-Floyd. These numbers are not only statistically insignificant but also small in magnitude.

Given the lack of evidence for targeting based on local demographics, we do not believe it drives our heterogeneous results. However, this highlights an untapped opportunity for lenders—to the extent that lenders can run local ads, they could increase their advertising effectiveness by tailoring ad content to local markets.

Table 6: Minority Representation in Ads and DMA Characteristics

	Minority Share in Ads			
	All ads (1)	Local ads (2)	All ads (3)	Local ads (4)
Minority Pop. Share	0.0103 (0.0196)	0.0619 (0.0610)		
<i>Dem</i>	-0.0096 (0.0153)	-0.0086 (0.0485)		
Pre-GF			0.2465*** (0.0047)	0.1955*** (0.0326)
Post-GF			0.3505*** (0.0048)	0.1530*** (0.0326)
Minority Pop. Share $\times$ Pre-GF			-0.0001 (0.0125)	0.0470 (0.0864)
<i>Dem</i> $\times$ Pre-GF			-0.0081 (0.0098)	-0.0991 (0.0690)
Minority Pop. Share $\times$ Post-GF			0.0049 (0.0125)	0.0794 (0.0850)
<i>Dem</i> $\times$ Post-GF			-0.0049 (0.0098)	0.0706 (0.0673)
Intercept	0.2958*** (0.0074)	0.1741*** (0.0232)		
<i>N</i>	404	391	404	391
<i>R</i> ²	0.0011	0.0030	0.7977	0.0252

Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

5.2.2 Targeting of minority representation based on channel viewership

Another potential concern with targeting is that lenders may place ads with higher minority representation on channels with more minority viewers. Since our ad spending measure does not vary among borrowers within a DMA, it does not account for potential differences in advertising exposure due to channel preferences across racial groups. Consequently, the observed greater responsiveness among minority consumers to ads with higher minority representation could be driven by greater exposure to such ads compared to White consumers.

To address this concern, we explore whether ads featuring more minority actors are indeed placed more frequently on channels with higher minority viewership. We combine TV channel-level ad spending data from Vivvix with channel viewership data by race from the MRI-Simmons Study, accessed through SimplyAnalytics.²⁰ The MRI-Simmons Study is based on a nationally representative sample of over 25,000 respondents and provides information on TV viewership by

²⁰<https://www.mrisimmons.com/partners/simply-analytics/>; accessed Dec. 2024

channel, broken out by race.

We construct viewership weights for White and minority consumers for each of the 102 major channels tracked by MRI-Simmons.²¹ Conceptually, the weight for White (or minority) consumers is calculated as the percentage of viewers in that group who watch a given channel divided by the overall percentage of viewers across both groups who watch that channel. More specifically, if $x\%$ of all consumers report watching a channel, and $x_w\%$ of White consumers and $x_m\%$ of minorities watch it, we define the White and minority weights as x_w/x and x_m/x , respectively. Thus, if minorities have higher viewership of a channel than White consumers, the minority weight is above 1, while the White weight is below 1. Across the 102 channels tracked in the data, the average weight for White consumers is 0.82 (std. dev. 0.15), and the average weight for minority consumers is 1.28 (std. dev. 0.30). These numbers align with broader statistics showing that minorities, particularly Black consumers, tend to spend more time watching TV compared to White consumers.²²

Using the combined data, we find little evidence that lenders target ads with higher minority representation to channels with greater minority viewership. Instead, the correlation between the minority share in ads and the minority viewership weight is negative (-0.078), when these two variables are weighted by the spending of each ad airing.²³ Moreover, descriptive patterns do not reveal clear trends. The average minority share in ads on the two channels with the highest minority viewership, TV ONE and BET, is 34% and 22%, respectively. For the channels with the highest White viewership, the History Channel and Fox News, the average minority shares are 28% and 40%, respectively.

Overall, we do not find strong empirical support for lenders strategically targeting ads with minority representation depending on local demographics or TV channels. That said, causal inference using observational data comes with inherent challenges because ads are not randomized by researchers. To complement the observational analysis, we conduct an experimental study where ads are directly manipulated and discuss the results in the next section.

²¹These channels account for 97.0% of total TV ad spending in our sample.

²²<https://www.statista.com/statistics/411806/average-daily-time-watching-tv-us-ethnicity/>; Accessed Feb. 6, 2025

²³When not weighted by ad spending (i.e., when each ad airing is counted equally regardless of spending amount), the correlation flips sign and becomes positive at 0.078.

6 Experimental Study

While we try to ensure that our findings are based on causal effects, it is always a challenge to ensure that estimated effects are causal based on observational data. Thus, we complement the observational data analysis with an experiment where we directly manipulate the race of actors using generative AI. The experimental study serves two purposes. First, despite having a rich set of fixed effects and video features as controls, one may still be concerned about endogeneity or omitted variable biases with observational data. An experiment with randomized ad assignments offers clean causal estimates. Although the magnitudes of results from an experiment and the field are not comparable, we can see whether our results pass external validity by seeing whether we get directionally similar results. Second, the experiment sheds light on potential mechanisms for our results through self-reported attitudinal questions we ask in the experiment. We describe the experimental design in Section 6.1 and discuss the results and implications in Section 6.2.

6.1 Experimental Design

We ran our experiment in June, 2023. As outlined in our preregistered research plan (https://aspredicted.org/B7C_P49), we aimed to recruit 2,800 participants through CloudResearch. We planned to recruit 2,000 mortgage borrowers or homeowners due to the limited pool of eligible participants, and supplement the remaining 800 subjects from the general population. We prioritized recruiting homeowners to ensure that the sample is comparable to the observational data. Ultimately, we ended up with 1,903 mortgage borrowers or homeowners and 902 general population participants. The difference between our intended subjects and our final subjects was that only 1,903 mortgage borrowers in the subject pool responded, and then we tried to get as close to the original 2,800 respondents as possible. After excluding 9 participants who failed the attention check,²⁴ the final sample consists of 2,796 participants with an average age of 42 years (excluding two outliers, 660 and 677, in the reported age) and 51.3% female. For racial composition, 72.4% of the participants identified as White/Caucasian, 8.7% as Black/African American, 8.1% as Asian, 3.6% as Hispanic, and the remaining 6.8% selected multiple or other categories. Finally, 10 participants (0.4%) chose not to disclose their race.

Participants were given the following description before seeing any stimuli: “In this survey, we would like you to imagine that you currently have a mortgage loan on your home and you

²⁴We asked the following question: Please select “2” for this question.

Figure 6: Example of Experimental Stimulus

AnchorPoint Refi's Advertisement:

Ad copy

Refinance with AnchorPoint Refi and start saving today! Whether you're looking to lower your monthly payments, consolidate debt, or fund home improvements, AnchorPoint Refi has got you covered.

Our customized solutions are tailored to your unique financial situation, and with our online application process, refinancing has never been easier. We'll help you get the best possible rate and save money with no stress or hassle.

Check out the people we've helped. Apply today at AnchorPointRefi.com and become our next success story!



AnchorPoint Refi is an Equal Housing Lender, committed to fair lending practices and licensed to operate in all 50 states. NMLS license number: #912328

Randomization

are considering refinancing the mortgage to reduce interest rates. You come across a refinance advertisement from AnchorPoint Refi. Please consider the ad as you are thinking about your refinancing decisions.” We then showed the respondents a print ad that is described below. We used print ads for this experiment because although our observational study focuses on TV ads, creating realistic videos that only vary the race of the actors is challenging. Instead, we created an artificial ad with a short description that was the same across conditions, and accompanied it with an image featuring customers, which was randomized across conditions. Figure 6 shows the stimuli with the short description and one of the images used in the study. The images were generated using Midjourney V5, a generative AI tool that creates highly realistic images from text prompts. To maintain consistency across images while varying only the race of the actors, we

provided Midjourney with detailed and explicit instructions.²⁵

Participants were randomly assigned to one of seven conditions. In each condition, the ad that was shown to the respondents had the same text and layout as the ad shown in Figure 6. The only thing that differed across conditions was the images shown at the bottom of the ads after the text. The 7 sets of pictures that were shown to the respondents appear in Figure 7. In four of the seven conditions, participants viewed an image featuring two families. As shown in Figure 7(a), these four conditions include two White families (WW), one White and one Black family (WB), one White and one Asian family (WA), or two Black families (BB). In all cases, the first family has two kids and the second family has one kid. The remaining three conditions, shown in Figure 7(b), featured an image with one White family (W), one Black family (B), or one Asian family (A). The purpose of having both two-family and single-family conditions is to explore whether the positive consumer responses observed in our observational study come from featuring racially diverse representation (two-family conditions) or just the presence of minorities in the ads (single-family conditions).

After participants were shown the ad, we measured two dependent variables (DVs): the likelihood of submitting a loan application with the advertised lender and the likelihood of recommending the lender to a friend who is looking to refinance. Both of these variables are measured on a 1-7 scale (1 = “Not at all likely”, 7 = “Very likely”). While the application likelihood aligns more closely with the DV in our observational analysis, recommendation likelihood may better reflect the overall brand attitude. The two measures are positively correlated ($\rho = 0.771$).

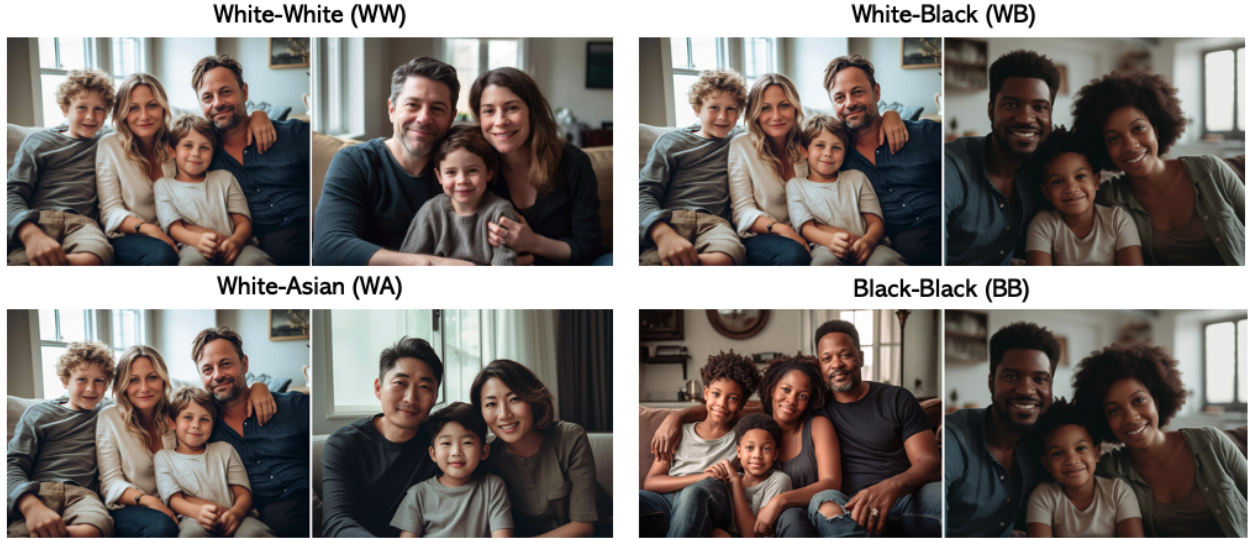
To explore potential mechanisms behind our results, we also asked participants to answer six attitudinal questions after providing their responses for the two DVs mentioned above. Using a 1-7 scale (1 = “Strongly disagree”, 7 = “Strongly agree”), the statements assess perceptions of product offerings (“The advertised lender has broad, flexible loan options that fit different financial situations and needs”), whether the respondent felt included (“The advertised lender caters to people like me”), financial inclusiveness (“The advertised lender is inclusive towards individuals of all backgrounds”), fair lending practices (“The advertised lender has fair lending practices without predatory interest rates and hidden fees”), the ad’s freshness (“The advertisement feels new and fresh”), and its ability to capture attention (“The advertisement is attention-grabbing”).

Lastly, we asked participants to provide their age, gender identity, and race and ethnicity

²⁵We used the prompt, “Good looking White dad and mom in their early 40s with two little kids, dressed casually, sitting on the couch, in the sunlit living room, white walls with windows in the background, No blur in the background, high detail, smiling and looking into the camera, photography –ar 4:3 –v 5 –s 700,” where the race and the number of kids varied across conditions as described below.

Figure 7: Advertising Images Featuring Different Races

(a) Two-family conditions



(b) Single-family conditions



(White/Caucasian, Black/African American, Hispanic/Latinx, Asian, Native American/Alaska Native, Mixed race/multiracial, Others, and Prefer not to disclose). They also reported their political orientation on a 1-7 scale (1 = “Very liberal”, 7 = “Very conservative”).

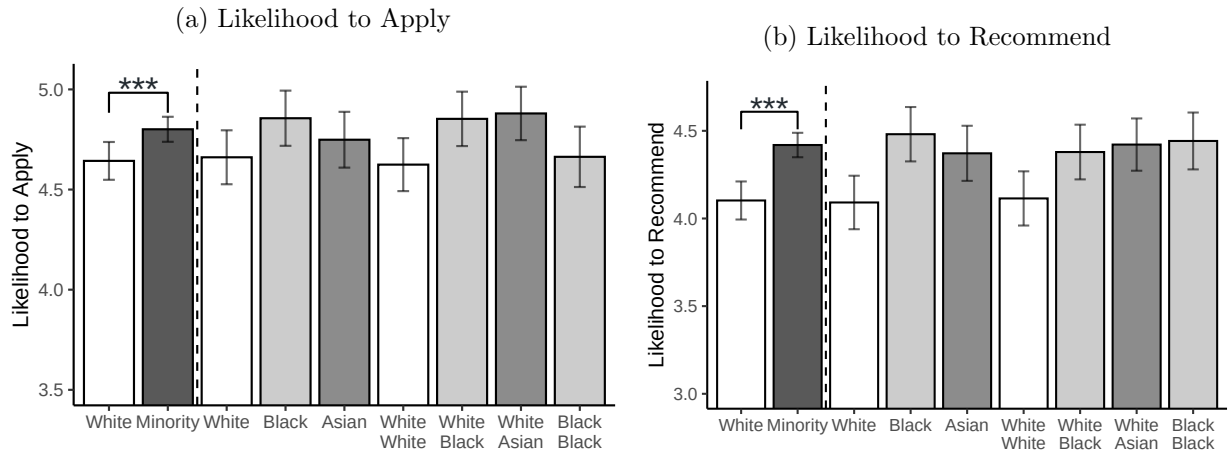
6.2 Experimental Results

The average response for the likelihood to apply appear in Figure 8(a) and the average results for the likelihood to recommend appear in Figure 8(b). On the left side of both figures, we see that ads featuring any minority families (combining WB, WA, BB, B, and A conditions) lead to a higher likelihood of application compared to those with only White families (combining WW

and W conditions) (4.80 vs. 4.64, $p = 0.006$). The effect is even stronger for the likelihood of recommending the lender (4.42 vs. 4.10, $p < 0.001$). These findings align with findings of the observational study that ads with more minority representation are more effective.

Figure 8 also presents results for each condition. Among the single-family conditions, participants in the Black condition report a higher likelihood to apply (4.86 vs. 4.66, $p = 0.048$) and recommend (4.48 vs. 4.09, $p < 0.001$) compared to the White condition. The Asian condition also shows a slightly higher application likelihood (4.75 vs. 4.66, $p = 0.376$) and a higher recommendation likelihood (4.37 vs. 4.09, $p = 0.012$) compared to the White condition. Among the two-family conditions, using two White families as the benchmark, participants in the White-Black condition report a higher likelihood to apply (4.85 vs. 4.62, $p = 0.018$) and recommend (4.38 vs. 4.11, $p = 0.018$). Similarly, in the White-Asian condition, the likelihood to apply (4.88 vs. 4.62, $p = 0.008$) and recommend (4.42 vs. 4.11, $p = 0.005$) are higher than in the two White families condition. The ad featuring two Black families shows a slight increase in the application likelihood (4.66 vs. 4.62, $p = 0.703$) and a significant increase in recommendation likelihood (4.44 vs. 4.11, $p = 0.004$) compared to ads with two White families. While the likelihood of applying is lower for the Black-Black condition than the White-Black condition, we do not believe that this shows a boundary condition that there may be an optimal level of minority presence because (1) this difference is statistically insignificant, and (2) we do not observe the same pattern for the likelihood of recommending the lender.

Figure 8: Likelihood to Apply and Recommend



Notes: The X-axis represents the race represented in the advertisement. The participant count for each condition: White ($n = 416$), Black ($n = 389$), Asian ($n = 382$), White-White ($n = 402$), White-Black ($n = 401$), White-Asian ($n = 408$), and Black-Black ($n = 398$). Error bars denote the 95% confidence interval. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

We also examine the heterogeneous effects based on participants' self-reported racial/ethnic information and political leaning. The results are in Table 7. The numbers represent the difference in likelihood to apply or recommend for ads featuring minority families (the five conditions combined) compared to ads featuring White families (WW and W). More detailed results for each condition separately are reported in Online Appendix F.

Table 7: Likelihood to Apply and Recommend by Race and Political Leaning

	Apply (1)	Recommend (2)	Apply (3)	Recommend (4)
Panel A: By Participants' Race				
	Minority Participants		White Participants	
Minority Ads	0.3110*** (0.1093)	0.3671*** (0.1238)	0.1006 (0.0692)	0.3033*** (0.0777)
<i>N</i>	761	761	2,025	2,025
Panel B: By Participants' Political Leaning				
	Liberal-leaning		Conservative-leaning	
Minority Ads	0.3979*** (0.0776)	0.5800*** (0.0887)	-0.1960* (0.1159)	-0.1384 (0.1280)
<i>N</i>	1,454	1,454	786	786
Panel C: By Participants' Race & Political Leaning				
	Minority Liberal		Minority Conservative	
Minority Ads	0.5279*** (0.1426)	0.5389*** (0.1639)	0.0021 (0.2438)	-0.0760 (0.2657)
<i>N</i>	423	423	169	169
	White Liberal		White Conservative	
Minority Ads	0.3391*** (0.0926)	0.5898*** (0.1056)	-0.2485* (0.1328)	-0.1297 (0.1471)
<i>N</i>	1,028	1,028	616	616
Notes: We omit reporting constants. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.				

Panel A shows that ads that feature minority actors significantly increase the likelihood of applying among minority participants, while the effect is positive but not significant among White participants. For the likelihood to recommend, both groups report a significantly higher likelihood when randomly assigned to ads featuring minority families compared to those featuring White families. Panel B reports the effects based on participants' self-reported political leanings. Liberal-leaning consumers (1-3 on a 1-7 scale) report a significantly higher likelihood to apply and to recommend when assigned to ads featuring minority families compared to ads with White families, while conservative-leaning consumers (5-7 on a 1-7 scale) show a slightly lower likelihood to apply and an insignificantly lower recommendation likelihood when assigned to ads featuring minority families. Finally, Panel C shows the interaction between race and political leaning. Among liberal-

leaning participants, the strong preference for ads featuring minority families persists regardless of their own race. No significant differences are observed in either measure for minority conservative participants. However, White conservative participants exhibit a modest decrease in the likelihood to apply when exposed to ads featuring minority actors.

Overall, the experimental results are broadly consistent with the observational study, confirming that the impact of minority representation is greater for minority consumers and liberal-leaning consumers. In the observational study, we rely on the census tract-level data to proxy political leaning, which inherently introduces measurement errors. The experimental results suggest that we would expect an even greater difference between liberal- and conservative-leaning consumers if our observational data used individual-level political ideology. In fact, the experimental results suggest that political leaning plays a more prominent role than race in shaping a customer’s response to ads featuring minority families.

7 Potential Mechanisms

In this section, we discuss several potential mechanisms that could explain our findings that minority representation increases the effectiveness of ads, as shown through both observational and experimental data. Our goal is not to pin down a single definitive mechanism; rather we show evidence of several possible explanations that are consistent with our findings. Indeed, the effect that ads with minority representation are more effective is likely to be multi-determined.

One potential explanation for our findings is that customers care about the racial match between themselves and the race of the people shown in the ads, with the importance of a match being especially high for minority customers. The past behavioral literature has found that racial fit plays a significant role in advertising effectiveness, particularly for minority consumers (e.g., [Deshpandé and Stayman, 1994](#); [Aaker, Brumbaugh, and Grier, 2000](#); [Mollica, Gray, and Trevino, 2003](#)). In our study, we find that minority consumers are significantly more likely to believe that “the advertised lender caters to people like me” when they see ads featuring minority actors, as opposed to ads with White actors (4.79 vs. 4.13, $p < 0.001$). White consumers, on the other hand, show only a marginal increase in the belief that ads featuring White actors cater to people like them compared to ads with minority actors (4.79 vs. 4.73, $p = 0.354$). Taking these two findings together, placing minority actors in advertisements should be more effective because of the positive effect this has on minority consumers and non-response from White consumers. This is consistent with our finding

that minority representation has a stronger impact on minority consumers from both observational and experimental studies (Table 4 and Table 7).

Racial fit is unlikely to be the only mechanism behind our results, however, since we find that ads that feature minority consumers are also more effective among White consumers, although at a smaller magnitude. There are a number of mechanisms that are consistent with this result, as well.

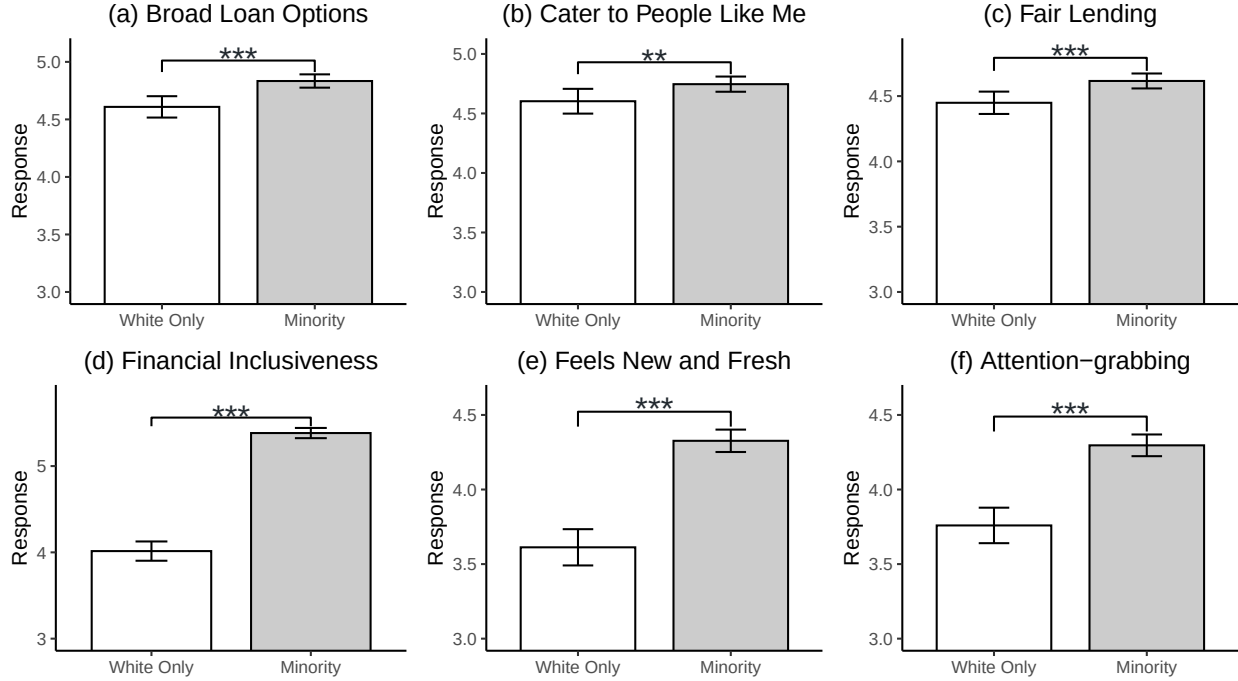
One possible mechanism is that the presence of minority actors in advertisements can speak to consumers’ support for diversity and minority representation. Several recent studies have documented greater support for DEI initiatives among liberal-leaning individuals (Agarwal and Sen 2022; Babar, Adeli, and Burtch 2023; Aneja, Luca, and Reshef 2025). Consistent with these studies, we find that liberal-leaning consumers, including White customers, respond more positively toward ads featuring minority actors compared to conservative-leaning consumers. The heterogeneity of the effect across the political spectrum suggests that the customer’s extent of support for diversity and minority representation, or their attitude about race in general, likely plays an important role in determining how consumers react to ads featuring minority consumers.

Moreover, the presence of minority actors in ads can affect consumers’ perceptions of the advertised brand. In other words, even when presented with the same ad copy for a fictional brand, consumers may perceive the brand differently based on the race of the actors. This aligns with prior research that finds that firms with diverse workforces are perceived to be more moral (Khan and Kalra, 2022). Figure 9 shows how brand perceptions vary for the six attitudinal measures from the experimental study according whether the participants were randomly assigned to see ads featuring White vs. minority families.²⁶ After seeing ads featuring minority consumers, participants are more likely to perceive the advertised lender as having broad and flexible loan options (4.83 vs. 4.61, $p < 0.001$), having fair lending practices without predatory pricing and hidden fees (4.62 vs. 4.45, $p = 0.001$), and being inclusive towards individuals of all backgrounds (5.38 vs. 4.01, $p < 0.001$). All these brand perceptions are positively correlated with the likelihood of application and recommendation (Table F3 in Online Appendix F). Minority representation can increase ad effectiveness through these favorable brand perceptions.

Lastly, ads featuring minority actors can be more effective simply because they are less common, making them stand out and appear more salient to consumers. Having minority representation in ads, therefore, can be an effective way for firms to differentiate their advertisements from others

²⁶More detailed results can be found in Online Appendix F

Figure 9: Consumer Perceptions of the Advertised Lender and Advertisement



Note: Error bars denote the 95% confidence interval; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and increase their effectiveness (Pieters, Warlop, and Wedel, 2002; Rosengren et al., 2020). In our experimental study, we find that participants are more likely to perceive the ads featuring minority actors as new and fresh (4.33 vs. 3.61, $p < 0.001$) and attention-grabbing (3.88 vs. 3.64, $p < 0.001$) compared to ads with White actors. Thus, we would expect that ads with minority actors would be more effective because of these visual cues, which are positively correlated with the likelihood of loan application and recommendation (Table F3 in Online Appendix F).

8 Conclusion

Given the growing emphasis on diversity and minority representation, it is crucial for brands to understand the impact of including racial minority actors in ads on advertising effectiveness. We find that greater minority representation in TV ads increases the effectiveness of advertising in the mortgage refinancing market. The impact of minority representation is stronger among minority consumers as well as liberal-leaning consumers. A pre-registered experimental study where we directly manipulate the race of the actors in ads shows consistent results with the observational study. We also discuss several potential mechanisms that are consistent with our findings.

Our research offers valuable insights for brands seeking to promote racially diverse and inclusive representation in their advertising strategies. Our results suggest that featuring minority actors not only achieves the social goal of increasing minority representation but also leads to higher advertising effectiveness, given the same level of advertising spending. Our results also have important policy implications. Since featuring minority actors in advertising is an effective strategy for reaching minority consumers, it has the potential to contribute to improving financial inclusion in the mortgage market. In particular, minority consumers have been shown to be less likely to refinance when it is beneficial to do so (Gerardi, Lambie-Hanson, and Willen, 2021; Gerardi, Willen, and Zhang, 2023). Increasing minority representation in TV advertising could encourage refinancing among minority consumers and help reduce racial disparities in the mortgage market.

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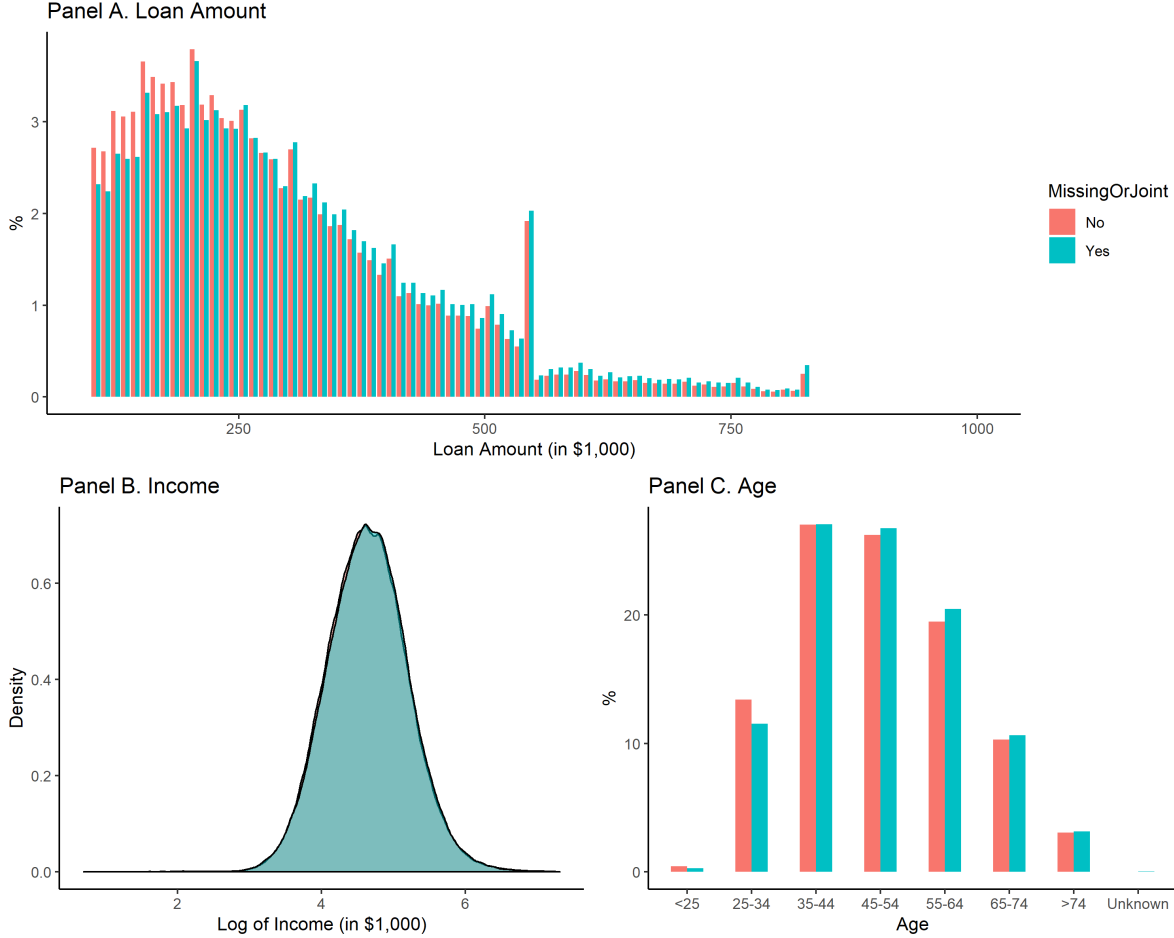
Online Appendix

A Data Cleaning Procedure

In this appendix, we outline our sample selection criteria and provide the remaining number of loans at each step. The data-cleaning procedure is as follows:

1. We begin with 42,841,482 mortgage loans originated by lenders for home purchase or refinance purposes from 2018 to 2021.
2. We exclude non-conventional, government-backed loans, including VA loans (for veterans and active-duty service members), FHA loans (for low-income and low-credit score borrowers), and USDA loans (for rural areas). We are left with 33,492,163 conventional loans.
3. Following prior literature (e.g., [Bartlett et al., 2022](#); [Bhutta, Hizmo, and Ringo, 2022](#); [Gerardi, Willen, and Zhang, 2023](#)), we apply additional inclusion criteria: loans must be first-lien mortgages for owner-occupied, site-built, single-family residential homes (excluding loans for multi-unit properties). Additionally, loans have a term between 10 to 30 years and a minimum loan size of \$100,000. After these filters, 24,696,189 loans remain.
4. We remove jumbo loans (loans exceeding conforming loan limits) and exclude “exotic loans” with the following characteristics: reverse mortgages, open-end credit, interest-only payments, non- or negative amortization, balloon payments, prepayment penalties, or loans with zero or over 20% APRs. We are left with 22,773,442 conforming loans.
5. We then select 13,920,602 refinanced mortgages by removing loans for home purchase.
6. For the purpose of studying TV advertising effectiveness, we further restrict the sample to loans where the underlying property is located in one of the top 101 Designated Market Areas (DMAs), the same as in the TV advertising data. We are left with 12,665,436 loans.
7. Since we will examine heterogeneous effects by borrower’s minority status, we exclude loans with missing ethnic/racial information or cases where borrowers on the same loan belong to different racial or ethnic backgrounds. Additionally, we remove a small number of loans originated to American Indian or Alaska Native and Native Hawaiian or Other Pacific Islander borrowers. To alleviate potential concerns about selection bias, [Figure A1](#) shows that the distributions of loan amount, borrower income, and age are similar between included and excluded loans.
8. Our final data sample consists of 9,668,300 refinance mortgage loans.

Figure A1: Loan Amount, Income, and Age for Loans with Missing/Joint Race Information



B Online Advertising

We start by comparing TV ad spending to that of online channels, including social media ads on Meta and online display ads. Meta ad data for the top 30 lenders is collected by inputting each lender's name into the Meta Ad Library,²⁷ where non-political ads is available for dates starting in May 2018. Online display ad data, like TV the ad data, is sourced from Vivvix. However, unlike TV ads, we do not have access to the ad creatives for online display ads. As a result, online display ad spending may include some irrelevant ads, such as those for reverse mortgages or HARP loans, which were removed from the TV ad data.

Table B1 reports lenders' average annual spending on Meta ads, online display ads, and TV ads. Meta ad spending is minimal. Spending on online display ads is more substantial but is still much lower compared to TV ads. Overall, smaller lenders (lenders are ordered by market share

²⁷<https://transparency.meta.com/researchtools/ad-library-tools>; Accessed Feb. 6, 2025

Table B1: Lenders’ Advertising Spending per Year by Channel

Lender	Meta ad spending (\$1,000s)	Online display ad spending (\$1,000s)	TV ad spending (\$1,000s)
Rocket Mortgage	27.11	19,354.64	200,895.54
United Wholesale Mortgage	—	366.15	44.35
Wells Fargo	—	481.07	6836.84
JP Morgan Chase	0.30	86.93	0.51
LoanDepot	3.30	1031.18	24,866.58
Nationstar	0.90	14.44	1428.21
Bank of America	—	45.02	—
Caliber Home Loans	0.01	83.87	—
US Bank	—	195.36	329.48
Fairway Independent	—	18.49	752.18
PennyMac	0.44	357.20	—
Guaranteed Rate	2.35	1191.19	7085.62
Flagstar Bank	—	25.91	37.06
Home Point Financial	—	28.84	—
Freedom Mortgage	0.01	1089.49	6.66
Newrez Mortgage	—	42.91	—
Provident Funding	—	—	—
AmeriSave Mortgage	0.31	324.15	7844.08
Better Mortgage	—	368.03	65.59
Citizens Bank	1.88	0.12	10.43
CrossCountry Mortgage	—	71.53	281.21
PNC Bank	—	9.13	0.70
Broker Solution	0.01	6.98	2.24
Cardinal Financial	0.44	6.67	5.71
Finance of America	—	11.25	15.66
Guild Mortgage	—	10.60	40.89
Fifth Third Bank	—	—	—
Huntington Natl. Bank	—	—	—
Movement Mortgage	0.03	19.15	1.13
American Financing	—	4.29	4200.46

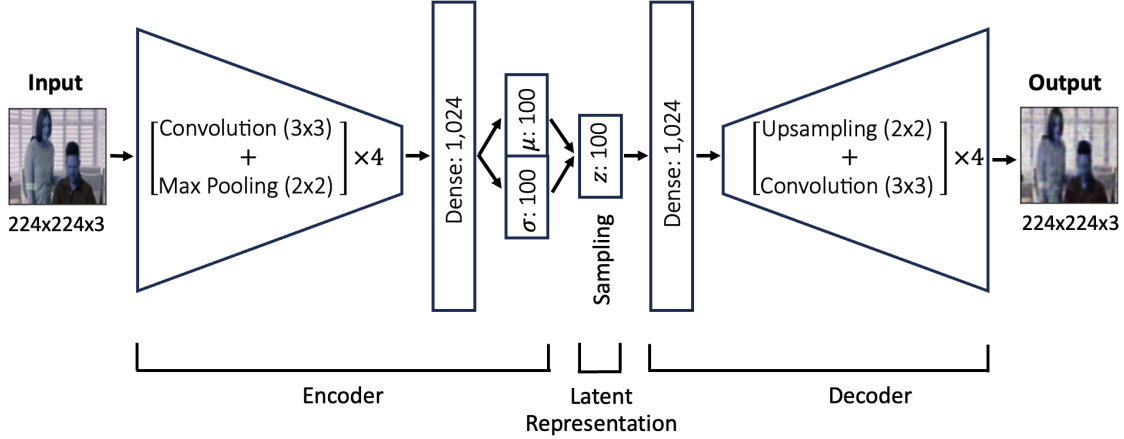
in the table) tend to allocate a higher proportion of their budget to online display ad spending relative to TV ads.

C VAE Architecture and Estimation Details

Figure C1 shows the structure of the Variational Autoencoder (VAE) used to represent image data in a lower-dimensional vector space, along with an example of an input image and the corresponding reconstructed image. As discussed in Section 3.1, an input image is represented as a 150,528-dimensional vector ($224 \times 224 \times 3$). This input is passed through the encoder network, which consists of four convolutional layers and two fully-connected/dense layers. In the convolutional layers, we apply commonly used 3×3 filters and 2×2 max pooling operations, gradually increasing the number of channels to 32, 32, 64, and 64. The purpose of this process is to extract meaningful

features from the image while reducing its dimensionality. The output from the convolutional layers is then flattened to a 12,544-dimensional vector ($14 \times 14 \times 64$) and passed through two fully-connected/dense layers. We use the Rectified Linear Unit (ReLU) as the activation function in these two layers. The output of the encoder network is the latent vector, denoted as z , which is sampled from a multivariate Gaussian distribution with parameters μ and σ . This latent vector has a dimensionality of 100, capturing a compact representation of the input image. The decoder network is the inverse of the encoder network. It takes the latent vector z as input and gradually increases its dimensionality until it matches the original vector space of the input image. The goal of the decoder network is to reconstruct an image close to the input image based on the compact representation z . The model is trained with a batch size of 64 for 100 epochs. We employ adaptive learning rates, starting with an initial rate of 0.005. We implement an early stopping rule to prevent overfitting.

Figure C1: VAE Network Structure

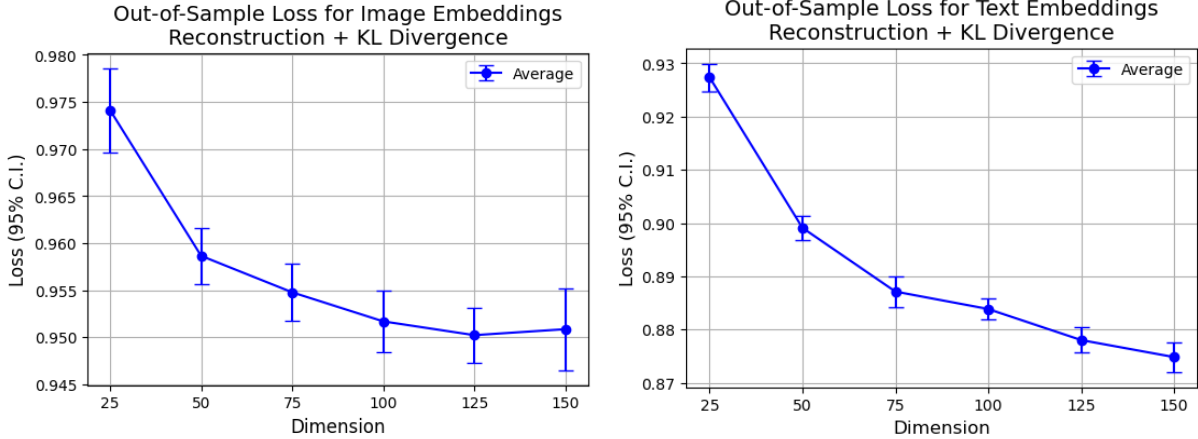


We implement another VAE to represent the 1,536-dimensional output of the OpenAI’s text embedding model (“text-embedding-ada-002”) into a 100-dimensional latent vector space. Since the input data does not contain spatial information unlike images, we use a simpler network structure that does not require convolutional neural networks. Specifically, we use fully connected layers for training. In the encoder network, we use a sequence of fully-connected layers with dimensions of 1024, 512, 256, and 100. In the decoder network, we perform the inverse operations, gradually increasing the dimensionality of the latent vector until it matches the original input space. The estimation procedure follows a similar approach as before. We train the model using a batch size of 64 for 100 epochs with adaptive learning rates that start with an initial rate of 0.005. We implement an early stopping rule to prevent overfitting.

We check the out-of-sample loss function across different latent embedding dimensions. Figure C2 illustrates the loss for both image (left figure) and text (right figure) data as the embedding

dimension increases from 25 to 150. The results suggest that for image data, the loss stabilizes at 100 dimensions. For text data, while the loss continues to decrease beyond 100 dimensions, the incremental improvement is relatively small.

Figure C2: Out-of-sample Loss with Different Embedding Dimensions



D Alternative Definition of Minority Share

We conduct two robustness checks using alternative definitions of minority share. In the first definition, we code the minority presence in an ad as 1 if there are any minorities in the ad and 0 otherwise. In the second definition, we code the minority presence in the ad as 1 if the minority share is above the median value in the data (23.5%) and 0 otherwise. We report the estimation results for both definitions in Table D1. Column (1) corresponds to whether there is an absence or presence of minority actors, while column (2) is based on whether the minority share is below or above the median minority share. Across both models, ads with a higher minority share are more effective, and the elasticities with respect to ad spending (specifically, for ads above certain levels of minority representation) are 0.0357 and 0.0210, respectively, which are of the same magnitudes as those in our main analysis. In contrast, the elasticities for ads below certain levels of minority representation are -0.0051 and 0.0053, respectively.

E Residual Variation in Advertising

One potential concern about controlling for a large set of fixed effects and high dimensional embeddings is whether we still have sufficient residual variation in advertising. Following the TV advertising literature (Shapiro, Hitsch, and Tuchman, 2021), we calculate the extent of the remaining variation in Ad and $Ad \times MS$ after accounting for the fixed effects and the embeddings.

More specifically, we first take the residual variation in Ad and $Ad \times MS$ that remains after training a neural network model, as described in Equation (6). The standard deviations of these

Table D1: Minority Representation in Ads and DMA Characteristics

	Lender choice	
	based on MS>0 (1)	based on MS>median value (2)
Ad - Low MS	-0.0228 (0.0174)	0.0106 (0.0105)
Ad - High MS	0.0551*** (0.0163)	0.0572*** (0.0155)
Num. of Unique Loans	9,668,300	9,668,300
Notes: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.		

residual variations are 0.00465 for Ad and 0.00217 for $Ad \times MS$. Next, we divide the standard deviations of the residual variation by the unconditional means of both variables, which are 0.02967 for Ad and 0.00969 for $Ad \times MS$. The ratios of the standard deviations to sample averages, or the coefficient of variation, are 0.157 for Ad and 0.224 for $Ad \times MS$. These values match the order of magnitude of those previously reported, such as 0.03 for the median across multiple CPG brands (Shapiro, Hitsch, and Tuchman, 2021) and 0.08 to 0.68 for auto insurance advertising (Tsai and Honka, 2021).

To evaluate whether the neural network model is properly trained, we compare its mean squared error (MSE) loss to that of a benchmark model that includes only fixed effects. For Ad , the neural network model’s MSE is approximately 47% of the benchmark model’s MSE, while for $Ad \times MS$, it is about 77%. We further test whether our neural network fully captures our intended fixed effects by regressing the residual variation from the neural network model on the fixed effects themselves. We find that the fixed effects do not add any additional explanatory power. These results suggest that the fixed effects are properly accounted for in model training, and that the video embeddings provide additional explanatory power beyond the fixed effects.

F Additional Results from Experiments

In this appendix, we present additional results from the experiment. Similar to Table 7 in the main text, we report the heterogeneous effects by participants’ race and political leaning. Instead of grouping the conditions into two groups, White ads and Minority ads, we report the results per condition. Specifically, we regress the likelihood of loan application and recommendation on the seven conditions, using the two White families condition as the baseline.

Table F1 presents the results. Panel A reports the results by participants’ race. Minority participants have a higher likelihood to apply and recommend for most of the conditions featuring minority actors compared to the baseline two-White families condition. White participants generally have a higher likelihood to recommend when exposed to minority ads. The effect on the likelihood to apply tends to be smaller and often not statistically significant. Panel B reports the

results by participants’ political ideology, defined the same way as described in Section 6.2. Among liberal-leaning participants, all conditions featuring minority families show a significantly higher likelihood to apply and recommend. Among conservative-leaning participants, the effects are generally negative but not statistically significant, except for the two Black families condition, which has a significant negative impact. Panel C reports the results by the interaction between race and political leaning. Among liberal participants, the effects are generally positive for all conditions featuring minority families, regardless of their own race. Among minority conservative participants, the effects are somewhat mixed and not statistically significant. For White conservative participants, the effects are generally negative, with a somewhat larger magnitude for application than for recommendation. Overall, these are all consistent with what we saw in Table 7 in the main text.

We further examine consumers’ perceptions of the advertised lender and advertisements across the seven conditions. Specifically, we regress each of the six attitudinal measures collected in our experiment on the seven experimental conditions, using the two White families as the baseline. The results are presented in Table F2. Overall, we observe that conditions featuring minority families lead to more positive perceptions across all dimensions we consider.

Lastly, we consider how the six attitudinal measures towards the lender and the advertisements correlate with the likelihood of loan application and recommendation. The results are presented in Table F3. In both Panel A and B, we observe that each of the six attitudinal measures is positively correlated with the likelihood to apply for the advertised lender and recommend it.

Table F1: Likelihood to Apply and Recommend by Race and Political Leaning per Condition

	Apply (1)		Recommend (2)		Apply (3)		Recommend (4)	
Panel A: By Participants' Race								
	Minority Participants				White Participants			
White-Black	0.4213**	(0.1844)	0.4366**	(0.2092)	0.1570	(0.1177)	0.1996	(0.1325)
White-Asian	0.3004	(0.1871)	0.2945	(0.2122)	0.2302**	(0.1164)	0.3101**	(0.1309)
Black-Black	0.3242*	(0.1840)	0.4677**	(0.2087)	-0.0739	(0.1181)	0.2746**	(0.1329)
White	0.1678	(0.1824)	0.0782	(0.2069)	-0.0192	(0.1167)	-0.0699	(0.1313)
Black	0.5600***	(0.1934)	0.4069*	(0.2193)	0.1029	(0.1170)	0.3409***	(0.1317)
Asian	0.3889**	(0.1862)	0.4184**	(0.2112)	0.0253	(0.1192)	0.2055	(0.1341)
N	761				2,025			
Panel B: By Participants' Political Leaning								
	Liberal-leaning				Conservative-leaning			
White-Black	0.3440***	(0.1320)	0.4358***	(0.1506)	-0.0611	(0.1936)	-0.0744	(0.2144)
White-Asian	0.4811***	(0.1285)	0.5473***	(0.1466)	-0.0957	(0.1941)	0.0957	(0.2148)
Black-Black	0.5155***	(0.1326)	0.8236***	(0.1513)	-0.5168***	(0.1945)	-0.3541	(0.2153)
White	0.0442	(0.1307)	0.0027	(0.1491)	-0.0693	(0.1949)	0.0289	(0.2158)
Black	0.3813***	(0.1334)	0.5863***	(0.1521)	-0.1484	(0.1945)	-0.1172	(0.2153)
Asian	0.3714***	(0.1315)	0.5237***	(0.1500)	-0.3496*	(0.2017)	-0.1802	(0.2233)
N	1,454				786			
Panel C: By Participants' Race & Political Leaning								
	Minority Liberal				White Liberal			
White-Black	0.3718	(0.2441)	0.4080	(0.2800)	0.3231**	(0.1571)	0.4293**	(0.1789)
White-Asian	0.4024*	(0.2333)	0.3810	(0.2676)	0.5224***	(0.1540)	0.6243***	(0.1753)
Black-Black	0.8530***	(0.2365)	0.9394***	(0.2713)	0.3614**	(0.1604)	0.7698***	(0.1827)
White	0.1360	(0.2401)	0.1828	(0.2754)	0.0068	(0.1558)	-0.0706	(0.1774)
Black	0.5738**	(0.2667)	0.5000	(0.3060)	0.2928*	(0.1550)	0.5912***	(0.1765)
Asian	0.7628***	(0.2374)	0.8744***	(0.2723)	0.1989	(0.1580)	0.3648**	(0.1799)
N	423				1,028			
	Minority Conservative				White Conservative			
White-Black	0.3425	(0.3695)	0.3846	(0.4045)	-0.2315	(0.2292)	-0.2243	(0.2545)
White-Asian	0.6282	(0.4251)	0.8846*	(0.4653)	-0.3005	(0.2242)	-0.0571	(0.2489)
Black-Black	-0.5718	(0.4103)	-0.5654	(0.4491)	-0.5697**	(0.2258)	-0.3149	(0.2507)
White	0.4009	(0.3977)	0.4301	(0.4354)	-0.2418	(0.2280)	-0.0892	(0.2531)
Black	0.5449	(0.3870)	-0.1154	(0.4236)	-0.3863*	(0.2280)	-0.1225	(0.2531)
Asian	-0.3718	(0.4251)	-0.2265	(0.4653)	-0.4072*	(0.2338)	-0.1756	(0.2595)
N	169				616			

Notes: We omit reporting intercepts. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table F2: Perceptions of the Advertised Lender and Advertisement

	(1)	(2)	(3)	(4)	(5)	(6)
	Broad Options	Cater to Me	Fair Lending	Inclusive	Fresh&New	Attention
White-Black	0.278*** (0.094)	0.258** (0.104)	0.200** (0.090)	1.857*** (0.100)	0.774*** (0.122)	0.658*** (0.118)
White-Asian	0.320*** (0.093)	0.350*** (0.103)	0.276*** (0.090)	1.716*** (0.099)	0.708*** (0.121)	0.610*** (0.118)
Black-Black	0.184** (0.094)	-0.030 (0.104)	0.099 (0.091)	1.421*** (0.100)	0.864*** (0.122)	0.612*** (0.118)
White	0.126 (0.093)	0.080 (0.103)	0.158* (0.090)	0.694*** (0.099)	0.187 (0.121)	0.182 (0.117)
Asian	0.272*** (0.095)	0.189* (0.105)	0.365*** (0.092)	1.823*** (0.101)	0.723*** (0.123)	0.451*** (0.120)
Black	0.391*** (0.094)	0.147 (0.104)	0.303*** (0.091)	1.793*** (0.101)	0.979*** (0.123)	0.814*** (0.119)
Intercept	4.545*** (0.066)	4.562*** (0.073)	4.368*** (0.064)	3.662*** (0.071)	3.517*** (0.086)	3.667*** (0.084)
N	2,796	2,796	2,796	2,796	2,796	2,796
Adj. R^2	0.006	0.005	0.006	0.179	0.035	0.023

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table F3: Impact of Consumer Perceptions on the Likelihood of Application and Recommendation

Panel A: Likelihood of Loan Application						
	Application					
	(1)	(2)	(3)	(4)	(5)	(6)
Broad Options	0.598*** (0.017)					
Cater to Me		0.591*** (0.014)				
Fair Lending			0.585*** (0.017)			
Inclusive				0.405*** (0.015)		
Fresh & New					0.453*** (0.013)	
Attention						0.452*** (0.013)
Intercept	1.906*** (0.082)	1.975*** (0.070)	2.081*** (0.083)	2.738*** (0.080)	2.888*** (0.056)	2.883*** (0.059)
<i>N</i>	2,796	2,796	2,796	2,796	2,796	2,796
Adj. <i>R</i> ²	0.318	0.381	0.286	0.202	0.320	0.296
Panel B: Likelihood of Recommendation						
	Recommendation					
	(1)	(2)	(3)	(4)	(5)	(6)
Broad Options	0.651*** (0.019)					
Cater to Me		0.632*** (0.017)				
Fair Lending			0.671*** (0.020)			
Inclusive				0.456*** (0.017)		
Fresh & New					0.538*** (0.014)	
Attention						0.552*** (0.014)
Intercept	1.223*** (0.094)	1.355*** (0.082)	1.263*** (0.093)	2.057*** (0.090)	2.111*** (0.062)	2.042*** (0.064)
<i>N</i>	2,796	2,796	2,796	2,796	2,796	2,796
Adj. <i>R</i> ²	0.297	0.342	0.295	0.200	0.353	0.347
Notes: * <i>p</i> < 0.1, ** <i>p</i> < 0.05, *** <i>p</i> < 0.01						